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Variational Monte Carlo for Droplets of Neutral Atom Mixtures

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Abstract

abstract

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Introduction

Solving the Schrödinger equation for interacting many-body systems is generally impossible analytically and standard numerical integration methods suffer from an exponential scaling in computational cost as the number of particles increases. To overcome this limitation, stochastic approaches such as Quantum Monte Carlo methods are widely employed to estimate the properties of complex quantum systems.

In this thesis, we investigate the thermodynamic and structural properties of a confined 3D bosonic neutral atom mixture. To this end, we develop a modular, Variational Monte Carlo engine capable of optimizing a Jastrow-correlated trial wavefunction. By tuning the cross-species Lennard-Jones interactions, we computationally explore the physics of the mixture, demonstrating the intricate competition between quantum kinetic pressure, short-range hard-core repulsion, and long-range attraction, which ultimately drives the system through a crossover from a trapped quantum gas to a self-bound quantum liquid droplet.

This thesis is the continuation and extension of the preceding Projects [1] and [2], and is structured as follows:

Chapter 1 : A comprehensive summary of the theory underpinning the Variational Monte Carlo method is presented. Section 1.4 specifically explores notable automated optimization algorithms tailored for energy minimization.

Chapter 2 : The physical system of interest is introduced and analyzed from an analytical perspective. Key mathematical quantities—crucial for the implementation of the chosen trial wavefunction—are explicitly derived, and relevant physical observables to be sampled via the Monte Carlo architecture are discussed.

Chapter 3 : An overview of the C++ code’s architecture, along with the core implementation choices and design patterns, is provided. The full source code can be found in the GitHub Repository at <https://github.com/DavideCacopardi/VMC-bachelor-thesis>.

Chapter 4 : The program is initially benchmarked against exactly solvable non-interacting scenarios. Subsequently, the numerical results for the interacting mixture are collected and thoroughly analyzed, uncovering the emergence of spatial clustering, phase separation, and the transition to the self-bound liquid regime.

Chapter 1

Variational Monte Carlo Fundamentals

1.1 Variational Principle

Consider a quantum mechanical system. Let the spectral decomposition of the Hamiltonian operator H be

$$H = \sum_n E_n |n\rangle\langle n|, \quad (1.1)$$

where $n \in \mathbb{N}$ and we sorted the energies in ascending order (i.e., $E_0 \leq E_1 \leq E_2 \leq \dots$) without loss of generality. We can consider a generic state $|\psi\rangle$ and write the expectation value

$$\langle\psi|H|\psi\rangle = \sum_n E_n \langle\psi|n\rangle \langle n|\psi\rangle. \quad (1.2)$$

A lower bound to the last expression is found exploiting the chosen sorting of the energies:

$$\langle\psi|H|\psi\rangle \geq E_0 \sum_n \langle\psi|n\rangle \langle n|\psi\rangle = E_0 \langle\psi|\psi\rangle. \quad (1.3)$$

We end up with the following well-known *variational principle*:

$$E_0 \leq \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}. \quad (1.4)$$

If the ground state $(E_0, |n\rangle)$ is unknown, one can take advantage of this inequality evaluating the upper-bound $\mathcal{E} = \langle\psi|H|\psi\rangle / \langle\psi|\psi\rangle$ by making use of an educated guess of the *trial state* $|\psi\rangle$. More specifically, one can vary the parameters which define $|\psi\rangle$ in order to find a tighter upper bound, closer to the true ground-state energy E_0 .

Working in the quantum D -dimensional coordinate-space of a system composed of N bodies, one can rewrite the expression for the estimator as

$$\mathcal{E} = \frac{\int d\mathbf{R} \psi^*(\mathbf{R}) H \psi(\mathbf{R})}{\int d\mathbf{R} |\psi(\mathbf{R})|^2}, \quad (1.5)$$

where we defined $\mathbf{R} = (\mathbf{r}_0, \dots, \mathbf{r}_{N-1})$ as the $(N \times D)$ -dimensional position vector of the particles building up our system. A generic Hamiltonian may be expressed in the following

form:

$$H = \sum_{i=0}^{N-1} \left[-\frac{\hbar^2}{2m_i} \nabla_i^2 + V_i(\mathbf{r}_i) \right] + \sum_{i=0}^{N-1} \sum_{j=0}^{i-1} U_{ij}(\mathbf{r}_i, \mathbf{r}_j) \quad (1.6)$$

1.2 Variational Monte Carlo

As previously suggested, the trial wavefunction employed can be defined by a set of M parameters $\boldsymbol{\alpha} = (\alpha_0, \dots, \alpha_{M-1})$:

$$\psi_T = \psi_T(\mathbf{R}; \boldsymbol{\alpha}).$$

We would like to explore different methods to optimize the estimator $\mathcal{E}$ with respect to the parameters $\boldsymbol{\alpha}$, but in order to do that we must first find a way to evaluate $\mathcal{E}$ efficiently.

1.2.1 Local Energy

We can rewrite Eq. 1.5 as follows:

$$\mathcal{E} = \frac{\int d\mathbf{R} \psi_T^* H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} = \frac{\int d\mathbf{R} \psi_T^* \frac{\psi_T}{\psi_T} H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} = \frac{\int d\mathbf{R} |\psi_T|^2 E_L}{\int d\mathbf{R} |\psi_T|^2} = \int d\mathbf{R} p(\mathbf{R}, \boldsymbol{\alpha}) E_L(\mathbf{R}, \boldsymbol{\alpha}), \quad (1.7)$$

where we defined the *local energy*:

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = \frac{1}{\psi_T(\mathbf{R}, \boldsymbol{\alpha})} H(\mathbf{R}) \psi_T(\mathbf{R}, \boldsymbol{\alpha}) \quad (1.8)$$

and the quantum PDF (*probability distribution function*):

$$p(\mathbf{R}, \boldsymbol{\alpha}) = \frac{|\psi_T(\mathbf{R}, \boldsymbol{\alpha})|^2}{\int d\mathbf{R}' |\psi_T(\mathbf{R}', \boldsymbol{\alpha})|^2}. \quad (1.9)$$

Evaluating the integral in equation 1.7 when $N \gg 1$ becomes computationally expensive if naive integration methods, such as trapezoidal integration, are employed. A better strategy is to use Monte Carlo integration methods instead.

1.2.2 The Vanishing-Variance Principle

A profound and extremely useful property of the local energy is known as the Zero-Variance Principle. Suppose our trial wavefunction $\psi_T(\mathbf{R})$ matches an exact eigenstate of the Hamiltonian, $\psi_0(\mathbf{R})$, with eigenvalue E_0 . In this scenario, the local energy evaluates to:

$$E_L(\mathbf{R}) = \frac{H\psi_0(\mathbf{R})}{\psi_0(\mathbf{R})} = \frac{E_0\psi_0(\mathbf{R})}{\psi_0(\mathbf{R})} = E_0$$

Remarkably, the local energy becomes completely independent of the spatial configuration $\mathbf{R}$. As a direct mathematical consequence, the statistical variance of the local energy, $\text{Var}(E_L)$, strictly vanishes as the trial wavefunction more finely approximates $\psi_0(\mathbf{R})$. Recent work by Webber and Lindsey [10] provides a rigorous proof for this quantitative extension, now formally referred to as the Vanishing-Variance Principle.

1.3 Monte Carlo Integration

For simplicity we rewrite Eq. 1.7, which contains the integral we need to evaluate.

$$\langle H \rangle = \int d\mathbf{R} p(\mathbf{R}, \alpha) E_L(\mathbf{R}, \alpha) \quad (1.10)$$

We can restrict ourselves to the 1-dimensional case and drop the dependency from the parameters α :

$$\mathbb{E}[E_L] = \int dx p(x) E_L(x) \quad (1.11)$$

If we draw a sufficiently large number N_{MC} of samples x_i from the distribution $p(x)$ we can also write:

$$\mathbb{E}[E_L] = \int dx p(x) E_L(x) \approx \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} E_L(x_i). \quad (1.12)$$

This step is justified by the *Strong Law of Large Numbers*, as long as we extract our points independently from one another.

Although this theoretical framework is elegant, it relies on a crucial assumption: our ability to extract independent samples directly from the complex, multidimensional probability distribution $p(\mathbf{R}) = |\psi_T(\mathbf{R})|^2 / \int d\mathbf{R}' |\psi_T(\mathbf{R}')|^2$.

In practice, generating independent random configurations for a quantum many-body system is computationally unfeasible. The probability density is typically sharply peaked in highly localized regions of the configuration space. If we were to naively sample the space using a uniform random distribution, the vast majority of our extracted points would land in regions where $p(\mathbf{R}) \approx 0$. These points would consume computational resources but contribute almost nothing to the final sum. Furthermore, as we increase the number of particles N , the volume of the configuration space grows exponentially.

To overcome this, we must leave aside the requirement of independent sampling. We use Markov Chain Monte Carlo methods to generate a sequence of *correlated* configurations that are mathematically guaranteed to asymptotically follow our target distribution $p(\mathbf{R})$.

1.3.1 Markov Chain

For simplicity, we consider a 1-dimensional system composed of one particle, whose state (e.g., its position) ranges over a discrete domain. Let $|\psi, t_0\rangle$ be its initial state, the probability of finding the system in a state i at a time $t_0 + \varepsilon$ is given by

$$|\langle i | \psi, t_0; t_0 + \varepsilon \rangle|^2 = w_i(t_0 + \varepsilon) = \sum_j W_{ij} w_j(t_0) \quad (1.13)$$

where the sum ranges over the whole domain of possible states j and $W_{ij} = W(j \rightarrow i)$ is an unknown transition probability from j to i . We are clearly assuming that W is time-independent. We can write the last equation in matrix form:

$$w(t_0 + \varepsilon) = W w(t_0) \quad (1.14)$$

and see that after n time steps

$$w(t_0 + n\varepsilon) = W^n w(t_0). \quad (1.15)$$

We can spectrally decompose the stochastic matrix W , obtaining eigenvalues λ_i , and expand the state vectors w in its diagonal basis $\{\nu_i\}$:

$$w(t_0 + n\varepsilon) = \sum_i \alpha_i \lambda_i^n \nu_i. \quad (1.16)$$

Since W is a stochastic matrix, it has the following properties:

$$\forall i, \quad \sum_j W_{ij} = 1 \quad \wedge \quad |\lambda_i| \leq 1 \quad \wedge \quad \max_i \{|\lambda_i|\} = 1 \quad (1.17)$$

This means we can sort the eigenpairs (λ_i, ν_i) so that

$$1 = |\lambda_0| \geq |\lambda_1| \geq |\lambda_2| \geq \dots \quad (1.18)$$

Now, we need to add the assumption that W is irreducible (i.e., starting from any state i , it is possible to reach any other state j). This allows the Perron-Frobenius Theorem to assure the multiplicity of $\lambda = 1$ to be exactly 1. Because of our sorting, $\lambda_0 = 1$. Furthermore, we require W to be aperiodic so that only λ_0 has magnitude 1.

$$1 = \lambda_0 > |\lambda_1| \geq |\lambda_2| \geq \dots \quad (1.19)$$

We can then rewrite equation 1.16 as

$$w(t_0 + n\varepsilon) = \alpha_0 \nu_0 + \alpha_1 \lambda_1^n \nu_1 + \alpha_2 \lambda_2^n \nu_2 + \dots \quad (1.20)$$

If we now take the limit for $n \rightarrow \infty$, we can see that the steady state $w = \alpha_0 \nu_0 = Ww$ is reached. After enough time steps, we are now allowed to rewrite Eq. 1.13 as

$$w_i = \sum_j W(j \rightarrow i) w_j. \quad (1.21)$$

$W(j \rightarrow i)$ is still unknown to us, but we can build a model for it. For instance we can write

$$w_i = \sum_j [w_j T(j \rightarrow i) A(j \rightarrow i) + w_i T(i \rightarrow j) (1 - A(i \rightarrow j))] \quad (1.22)$$

where T are the probabilities of proposing a certain transition and A are the probabilities of accepting said transition; the two terms in the sum account both for the probability of *transitioning to* state i and for the probability of *staying in* state i (i.e., *not transitioning elsewhere*). Simplifying the equation leaves us with

$$\sum_j w_j T(j \rightarrow i) A(j \rightarrow i) = \sum_j w_i T(i \rightarrow j) A(i \rightarrow j) \quad (1.23)$$

which is satisfied by a stronger assumption referred to as *detailed balance*:

$$w_j T(j \rightarrow i) A(j \rightarrow i) = w_i T(i \rightarrow j) A(i \rightarrow j) \quad (1.24)$$

We can now return to the more general configuration space in 3 dimensions where the system of N particles lives.

In the end, we have come to know

$$\frac{T(j \rightarrow i) A(j \rightarrow i)}{T(i \rightarrow j) A(i \rightarrow j)} = \frac{w_i}{w_j} = \frac{p(\mathbf{R}_i)}{p(\mathbf{R}_j)} = \frac{|\psi_T(\mathbf{R}_i)|^2}{\int d\mathbf{R} |\psi_T(\mathbf{R})|^2} \frac{\int d\mathbf{R} |\psi_T(\mathbf{R})|^2}{|\psi_T(\mathbf{R}_j)|^2} = \frac{|\psi_T(\mathbf{R}_i)|^2}{|\psi_T(\mathbf{R}_j)|^2} \quad (1.25)$$

and now we can choose different approaches.

1.3.2 Brute Force Metropolis Sampling

The simplest method to construct this proper Markov chain is the standard random-walk Metropolis algorithm we often refer to as "brute force" Variational Monte Carlo. In this approach, we start from an arbitrary initial configuration $\mathbf{R}$ and iteratively propose new configurations $\mathbf{R}'$ by shifting a single particle's coordinates by a random amount. Specifically, if a particle is currently at position $\mathbf{r}_i$, we propose a new position $\mathbf{r}'_i$ using a uniform random displacement:

$$\mathbf{r}'_i = \mathbf{r}_i + \delta \cdot (\mathbf{u} - 0.5) \quad (1.26)$$

where $\mathbf{u}$ is a vector of random numbers uniformly distributed in $[0, 1]$, and δ is a set maximum step length. Since this proposal method is perfectly symmetric, $T(\mathbf{R} \rightarrow \mathbf{R}') = T(\mathbf{R}' \rightarrow \mathbf{R})$ which simplifies the detailed balance equation down to:

$$\frac{A(\mathbf{R} \rightarrow \mathbf{R}')}{A(\mathbf{R}' \rightarrow \mathbf{R})} = \frac{p(\mathbf{R}')}{p(\mathbf{R})} \quad (1.27)$$

This places a constraint on the ratio of the acceptance probabilities, but it does not uniquely define them. To maximize the efficiency of the configuration-space exploration, we want to accept proposed moves as frequently as possible when they take us to regions of greater interest. Thus, we choose the largest possible values for A that strictly remain valid probabilities ($A \leq 1$).

If the proposed state is more probable than the current state ($p(\mathbf{R}') \geq p(\mathbf{R})$), we automatically accept the move by setting $A(\mathbf{R} \rightarrow \mathbf{R}') = 1$. If the proposed state is less probable, we set the reverse acceptance $A(\mathbf{R}' \rightarrow \mathbf{R}) = 1$, which forces the forward acceptance to exactly equal the probability ratio. Summing up:

$$A(\mathbf{R} \rightarrow \mathbf{R}') = \min \left(1, \frac{p(\mathbf{R}')}{p(\mathbf{R})} \right) \quad (1.28)$$

Algorithmically, at each step, we extract a uniform random number $u' \in [0, 1]$. If $u' \leq A$, the move is accepted and $\mathbf{R}'$ is added to the chain. If $u' > A$, the move is rejected, and the old configuration $\mathbf{R}$ is counted again.

While computationally cheap per step, this symmetric random walk is physically "blind". The acceptance ratio will heavily depend on the chosen step length parameter δ . Setting δ too high increases the probability of the proposed move being rejected because moves will likely be proposed into regions where the wavefunction approaches zero. On the other hand, opting for a smaller step length leads to highly correlated samples and a long thermalization time. Thus the standard trade-off is to tune δ so as to achieve a $\sim 50\%$ acceptance ratio, which still means that half of the proposal steps are "wasted".

1.3.3 Importance Sampling and Fokker-Planck Dynamics

To drastically improve the sampling efficiency, we introduce *Importance Sampling*. Instead of a blind random walk, we model the particles as undergoing diffusion driven by a drift force, governed by the Langevin equation.

We introduce a *quantum force* $\mathbf{F}_Q$, which actively pushes the particles towards regions where the probability density $|\psi_T|^2$ is maximized. The quantum force is defined as the logarithmic gradient of the probability density:

$$\mathbf{F}_Q(\mathbf{R}) = 2 \frac{\nabla \psi_T(\mathbf{R})}{\psi_T(\mathbf{R})} = 2 \nabla \ln(\psi_T(\mathbf{R})) \quad (1.29)$$

Using this force, the proposed move for a particle is no longer symmetric. It is updated using a normally distributed random variable and a deterministic drift ($\mathbf{v}_{\text{drift}}(\mathbf{r}_i) = \mathbf{F}_Q(\mathbf{r}_i)$):

$$\mathbf{r}'_i = \mathbf{r}_i + D\mathbf{v}_{\text{drift}}(\mathbf{r}_i)\Delta t + \boldsymbol{\eta}\sqrt{2D\Delta t} \quad (1.30)$$

where $D = \hbar^2/(2m)$ is the diffusion constant, Δt is the simulated time step, and $\boldsymbol{\eta}$ is a random vector drawn from a standard normal distribution $\mathcal{N}(\mu = 0, \sigma = 1)$.

Because the proposal is no longer symmetric, we must use the generalized *Metropolis-Hastings* acceptance rule to preserve detailed balance. We must explicitly include the Green's function $G(\mathbf{R} \rightarrow \mathbf{R}')$, which defines the transition probability of the Langevin equation:

$$G(\mathbf{R} \rightarrow \mathbf{R}') = \frac{1}{(4\pi D\Delta t)^{3N/2}} \exp\left(-\frac{(\mathbf{R}' - \mathbf{R} - D\mathbf{v}_{\text{drift}}(\mathbf{R})\Delta t)^2}{4D\Delta t}\right) \quad (1.31)$$

The new acceptance ratio A becomes:

$$A(\mathbf{R} \rightarrow \mathbf{R}') = \min\left(1, \frac{G(\mathbf{R}' \rightarrow \mathbf{R})}{G(\mathbf{R} \rightarrow \mathbf{R}')} \frac{|\psi_T(\mathbf{R}')|^2}{|\psi_T(\mathbf{R})|^2}\right) \quad (1.32)$$

This directed sampling approach ensures that large steps can be taken in the configuration space while maintaining a high acceptance ratio, significantly reducing the autocorrelation time of the Markov chain.

1.3.4 Umrigar Drift

If the probability distribution contains singularities, the quantities measured by employing the standard Monte Carlo techniques discussed until now may be harshly biased by the walkers never sampling the neighborhoods of singularities or, conversely, starkly disturbed by them in the rare scenario of them being sampled.

Because of this, one can apply a small correction to the formerly laid out math: as proposed in Umrigar's work [9], the drift term of Eq. 1.30 can be modified into

$$\mathbf{v}_{\text{drift}}(\mathbf{r}_i) = \frac{\sqrt{1 + 2a|\mathbf{F}_Q(\mathbf{r}_i)|^2\Delta t} - 1}{a|\mathbf{F}_Q(\mathbf{r}_i)|^2\Delta t} \mathbf{F}_Q(\mathbf{r}_i), \quad (1.33)$$

where $a \in [0, 1]$ is an interpolation parameter. As noted by Umrigar [9], the standard drift relies on the assumption that the quantum force $\mathbf{F}_Q = 2\nabla\psi_T/\psi_T$ is constant over the time step. Near a node, however, it is much more accurate to assume that $\nabla\psi_T$ is constant instead. The parameter a interpolates between these two regimes, effectively bounding the displacement and preventing instabilities when $|\mathbf{F}_Q| \rightarrow \infty$.

1.4 Optimization of Parameters

So far we have studied how we can evaluate the expectation value $\langle H \rangle$, given a fixed parameter vector $\boldsymbol{\alpha}$. This means we can list an array of reasonable $\boldsymbol{\alpha}$, let our program run for each one of them and estimate the optimal parameter vector (i.e., the one that minimizes the energy $E = \langle H \rangle$) visually, by means of a grid-search plot. Naturally, this serves only as a rough estimate and becomes increasingly more difficult and unreliable when our parameter vector has more and more components.

Moreover, since E is evaluated stochastically, optimizing it directly can sometimes lead to numerical instabilities: an artificially lower mean energy might be achieved inadvertently pushing the parameters into regions where the wavefunction exhibits unphysical behavior and the variance drastically explodes.

1.4.1 The Loss Function

Recalling the Vanishing-Variance Principle (Subsection 1.2.2), we know that the true ground state concurrently minimizes the energy and strictly yields a zero variance for E_L . To mitigate the previously mentioned instabilities, it is highly advantageous to guide our search optimizing a mixed loss function that reflects both physical requirements:

$$\mathcal{L}(\boldsymbol{\alpha}) = \langle E_L \rangle + w \text{Var}(E_L),$$

where w is a tunable weight parameter.

This formulation acts as a physical regularizer. The $\langle E_L \rangle$ term naturally drives the parameters toward the minimum energy, while the $w \text{Var}(E_L)$ penalty heavily discourages trial wavefunctions that produce wildly fluctuating local energies. The overall result is a much smoother optimization landscape, which prevents from overfitting the stochastic noise and efficiently guides the search toward a trial state that closely mimics the true target eigenstate.

It is important to point out that the chosen functional form for the trial wavefunction is fundamentally restricted and might not be able to represent the true ground state exactly. Because of this analytical limitation, the set of parameters that minimizes the expected energy does not generally coincide with the one that minimizes the variance. The introduction of the weight w thus represents a deliberate trade-off between achieving the lowest possible variational energy and obtaining a smoother, computationally more stable local energy profile.

Once the loss function to optimize is established, one can abandon the manual grid search and make use of automated optimization algorithms. A brief overview of the selected methods follows.

1.4.2 Newton-Raphson's Algorithm

Newton-Raphson's algorithm starts off by fixing an initial educated guess for the parameter vector. Let $\mathbf{g}(\boldsymbol{\alpha}) := \nabla_{\boldsymbol{\alpha}} \mathcal{L}(\boldsymbol{\alpha})$. At any n -th guess $\boldsymbol{\alpha}_n$ for the parameter vector, we can Taylor-expand $\mathcal{L}(\boldsymbol{\alpha})$ in the neighborhood of $\boldsymbol{\alpha}_n$ and evaluate it at $\boldsymbol{\alpha} = \hat{\boldsymbol{\alpha}}$, the optimal parameter vector we are seeking.

$$\begin{aligned} \mathcal{L}(\hat{\boldsymbol{\alpha}}) &= \mathcal{L}(\boldsymbol{\alpha}_n) + \mathbf{g}_n^\top (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n) + \frac{1}{2} (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n)^\top A_n (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n) + O((\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n)^3) = \\ &= \mathcal{L}_n + \mathbf{g}_n^\top \mathbf{b}_n + \frac{1}{2} \mathbf{b}_n^\top A_n \mathbf{b}_n + O(|\mathbf{b}_n|^3) \end{aligned} \tag{1.34}$$

where we defined

$$\mathcal{L}_n := \mathcal{L}(\boldsymbol{\alpha}_n), \quad \mathbf{b}_n := \hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n, \quad \mathbf{g}_n := \mathbf{g}(\boldsymbol{\alpha}_n), \quad A_n := \left\{ \partial_{\alpha_i \alpha_j}^2 \mathcal{L} \right\}_{i,j} \bigg|_{\boldsymbol{\alpha}_n}.$$

Now we differentiate both sides and set them to zero. By doing so, we find $\mathbf{b}_n = -A_n^{-1} \mathbf{g}_n$, which is Newton-Raphson's iterative proposal:

$$\boldsymbol{\alpha}_{n+1} = \boldsymbol{\alpha}_n - A_n^{-1} \mathbf{g}_n. \tag{1.35}$$

At this point we could iterate this method until $\|\alpha_{n+1} - \alpha_n\|_2 \leq \varepsilon$, where ε is a chosen precision (typically $\sim 10^{-5}$).

The enormous issue with this process is that evaluating the exact A_n^{-1} matrix is extremely expensive. Thus, we are interested in finding a way to estimate it; that is, we rewrite the proposal to the more general expression

$$\alpha_{n+1} = \alpha_n - \eta_n C_n \mathbf{g}_n, \quad (1.36)$$

where η_n is a scalar *learning rate*, and try to find a matrix C_n that well approximates the inverse Hessian A_n^{-1} while remaining computationally cheap.

1.4.3 Broyden-Fletcher-Goldfarb-Shanno (BFGS) Algorithm

Starting off with a principle similar to the one used in the one-dimensional secant method, the BFGS algorithm iteratively updates C_n without needing costly matrix inversions. Defining the parameter step $\mathbf{s}_n := \alpha_{n+1} - \alpha_n$ and the gradient change $\mathbf{y}_n := \mathbf{g}_{n+1} - \mathbf{g}_n$, the update rule for the inverse Hessian is given by:

$$C_{n+1} = \left(\mathbb{1} - \frac{\mathbf{s}_n \mathbf{y}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n} \right) C_n \left(\mathbb{1} - \frac{\mathbf{y}_n \mathbf{s}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n} \right) + \frac{\mathbf{s}_n \mathbf{s}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n}. \quad (1.37)$$

Note that the underlying convergence proof requires the loss function $\mathcal{L}(\alpha)$ to be convex; this condition is usually satisfied in a neighborhood of the local minimum for cases of major interest.

1.4.4 Adaptive Moment Estimation (Adam) Algorithm

In this optimization algorithm [7], running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. It is the method of choice for most neural network trainings, as it is simple enough, highly efficient even in higher dimensions and sufficiently noise-forgiving for most applications. It still requires a lot of experimenting to find the *learning rate* parameter η that works the best in each specific scenario. Starting off by fixing an initial educated guess for the parameter vector, the iteratively proposed parameter update is

$$\alpha^{(t)} := \alpha^{(t-1)} - \eta \frac{\hat{m}_\alpha^{(t)}}{\sqrt{\hat{v}_\alpha^{(t)} + \varepsilon}}, \quad (1.38)$$

where $t = 1, 2, 3, \dots$ is the iteration index, $\varepsilon \ll 1$ prevents numerical errors, and

$$\hat{m}_\alpha^{(t)} = \frac{1}{1 - \beta_1^t} \left[\beta_1 m_\alpha^{(t-1)} + (1 - \beta_1) \nabla_\alpha \mathcal{L}^{(t-1)} \right],$$

$$\hat{v}_\alpha^{(t)} = \frac{1}{1 - \beta_2^t} \left[\beta_2 v_\alpha^{(t-1)} + (1 - \beta_2) \left(\nabla_\alpha \mathcal{L}^{(t-1)} \right)^2 \right]$$

are the first and second moment.

1.4.5 Statistical Energy Gradient and Variance Gradient

In order to feed any gradient-descent based optimizer, such as BFGS or Adam, we must be able to efficiently compute the objective gradient $\mathbf{g}(\boldsymbol{\alpha}) = \nabla_{\boldsymbol{\alpha}} \mathcal{L}$.

Evaluating the derivative of the expectation value of the energy analytically yields:

$$\frac{\partial}{\partial \alpha_i} \langle E_L \rangle = \frac{\partial}{\partial \alpha_i} \left(\frac{\int d\mathbf{R} \psi_T^* H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} \right) = 2 \left(\left\langle E_L \frac{\partial \ln |\psi_T|}{\partial \alpha_i} \right\rangle - \langle E_L \rangle \left\langle \frac{\partial \ln |\psi_T|}{\partial \alpha_i} \right\rangle \right) \quad (1.39)$$

We can rewrite it introducing the operator $O_{\alpha_i}(x) = \frac{\partial \ln |\psi_T|}{\partial \alpha_i}$:

$$\frac{\partial}{\partial \alpha_i} \langle E_L \rangle = 2(\langle O_{\alpha_i} E_L \rangle - \langle O_{\alpha_i} \rangle \langle E_L \rangle) = 2 \text{Cov}(O_{\alpha_i}, E_L) \quad (1.40)$$

Operating in an analogous way, we can find the following gradient of the variance:

$$\frac{\partial}{\partial \alpha_i} \text{Var}(E_L) = 2 \text{Cov}(O_{\alpha_i}, E_L^2) - 4 \langle E_L \rangle \text{Cov}(O_{\alpha_i}, E_L) \quad (1.41)$$

This means that we can evaluate $\mathbf{g} = \nabla_{\boldsymbol{\alpha}} \langle E_L \rangle + w \nabla_{\boldsymbol{\alpha}} \text{Var}(E_L)$ simultaneously during the same Monte Carlo process used to compute the energy itself.

We simply need to sample the logarithmic derivative of the trial wave function alongside the local energy.

Unfortunately, this covariance evaluation is notoriously susceptible to high levels of stochastic noise and this may severely disrupt gradient-based algorithms (especially quasi-Newton methods like BFGS that approximate the Hessian matrix).

1.4.6 Nelder-Mead Algorithm

To circumvent the noise instability of the gradients, one can resort to derivative-free optimization algorithms, such as the Nelder-Mead method. Relying strictly on direct point-to-point comparisons of the loss function, this algorithm operates on an observable that inherently possesses a much higher signal-to-noise ratio than its gradient.

Although Nelder-Mead is widely successful in practical applications, it is fundamentally a heuristic algorithm based on geometric transformations of a simplex. In fact, rigorous mathematical proofs of its convergence only exist for strictly convex functions in one and two dimensions, while counterexamples demonstrating non-convergence are well known for higher dimensions. Furthermore, as the simplex shrinks near the minimum, standard Nelder-Mead can still be trapped by the residual statistical fluctuations of the evaluated energy. Therefore, specific implementational strategies—which will be discussed in Chapter 3—must be adopted to dynamically mitigate the noise and ensure meaningful convergence.

1.5 Error Analysis

Once we have found the optimal parameters, we want to establish a consistent way to assign uncertainties to the energy expectations evaluated with our Monte Carlo algorithm.

We do not want to run multiple independent Monte Carlo simulations from scratch because that would be computationally prohibitive. Instead, we can extract a good estimate of the true statistical error from a single long run using resampling techniques. While methods like the *bootstrap* are conceptually simpler to implement, the *Blocking* method is specifically

designed to rigorously account for the non-zero covariance of our data-set.

As previously noted, Markov Chains correlate consecutive Monte Carlo steps. Consequently, the naive standard error of the mean underestimates the true statistical uncertainty. Evaluating the exact covariance requires computing the auto-correlation function, which implies a double-loop over the dataset resulting in an $\mathcal{O}(n^2)$ complexity (where $n \sim 10^6$ is the number of Monte Carlo steps); this is unfeasible. However, we can exploit the exponential decay of the auto-correlation function, which is a direct consequence of the Markov property (where the proposed step depends only on the current state).

This is the founding principle for the blocking algorithm.

1.5.1 The Automated Blocking Algorithm

Flyvbjerg and Petersen popularized the "blocking method" [4], which involves applying a "blocking transformation" to the initially correlated data points. This blocking transformation consists of grouping consecutive data points into larger "blocks" and forming a new, smaller data-set composed of the mean of each block.

For instance, let our original dataset of local energies be $X^{(0)}$ of size n . In the first blocking transformation, we average adjacent data points to create a new dataset $X^{(1)}$ of size $n/2$:

$$X_i^{(1)} = \frac{1}{2} \left(X_{2i}^{(0)} + X_{2i+1}^{(0)} \right) \quad (1.42)$$

This halving process can be repeated up to $d = \log_2(n)$ times.

As the block size increases, the new block variables $X^{(k)}$ become increasingly independent. If we compute the apparent variance of the sample mean at each blocking step k , we observe that it grows and eventually plateaus at the *true* variance of the mean, at which point the block size is larger than the autocorrelation time. Once the plateau is reached, the computed apparent variance starts oscillating greatly, mainly because the sample size has become too small and thus the statistical uncertainty of the variance itself has rapidly increased.

Marius Jonsson's work [6] provides a rigorous mathematical proof of this method and also an automated routine to determine which blocking step is more likely to be the start of the plateau, within a set confidence level. This ultimately provides a rapid, effective and unsupervised blocking estimate.

Chapter 2

The Physical System: Neutral Atom Mixtures

2.1 The Physical System

The physical system we want to study is a mixture of N neutral atoms. All atoms are assumed to have the same mass m . Each atom is characterized by a *flavor*, which can be either A or B . The number of A atoms is always equal to the number of B atoms; that is, $N_A = N_B = N/2$ and thus the total number of atoms N is always even.

The mixture is subject to two different kinds of potential: a harmonic trap and Lennard-Jones interactions. In other words, in coordinate space the physical system is described by the following Hamiltonian:

$$H = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \nabla_i^2 + \frac{1}{2} \omega^2 \sum_{i=0}^{N-1} r_i^2 + \sum_{i < j} U_{m,n}(r_{ij}), \quad (2.1)$$

where $U_{m,n}(r_{ij})$ is the Lennard-Jones interaction term

$$U_{m,n}(r_{ij}) = 4\epsilon_{m,n} \left[\left(\frac{\sigma_{m,n}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{m,n}}{r_{ij}} \right)^6 \right], \quad (2.2)$$

acting on the i -th particle, whose flavor is m , and the j -th particle, whose flavor is n . For simplicity, we set the following equalities:

$$\epsilon = \epsilon_{A,A} = \epsilon_{B,B},$$

$$\epsilon_{A,B} = \tilde{\alpha}\epsilon,$$

$$\sigma = \sigma_{A,A} = \sigma_{B,B} = \sigma_{A,B}.$$

This means pairs of particles alike in flavor ((A, A) , (B, B)) interact according to

$$U_{AA}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (2.3)$$

while pairs of particles of different flavors (A, B) interact according to

$$U_{AB}(r_{ij}) = 4\tilde{\alpha}\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right]. \quad (2.4)$$

Furthermore, to fix the energy and length scales of our system, we set:

$$\frac{\hbar^2}{2m} = \frac{\epsilon\sigma^2}{10}. \quad (2.5)$$

2.2 Trial Wavefunction

To exploit the variational principle (Section 1.1), we choose the following trial wavefunction:

$$\psi(\mathbf{R}; \boldsymbol{\alpha}) = \prod_i e^{-\frac{r_i^2}{2\alpha}} \prod_{i<j} e^{-\frac{1}{2}u_\beta(r_{ij})} = \exp \left(-\frac{1}{2\alpha} \sum_i r_i^2 - \frac{1}{2} \sum_{i<j} u(r_{ij}; \beta, \gamma) \right), \quad (2.6)$$

where the Jastrow terms have the form

$$u(r_{ij}; \beta, \gamma) = \left(\frac{\beta}{r_{ij}} \right)^\gamma. \quad (2.7)$$

To avoid relying on numerical differentiation—which significantly increases the computational cost and introduces numerical artifacts—it is highly advantageous to analytically derive the quantities required for the Variational Monte Carlo implementation.

2.2.1 Local energy

The central quantity we need to evaluate is local energy, as previously explained in Section 1.2.1. From Eq. 1.8 we have

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = \frac{1}{\psi(\mathbf{R}, \boldsymbol{\alpha})} H(\mathbf{R}) \psi(\mathbf{R}, \boldsymbol{\alpha}). \quad (2.8)$$

Plugging in the Hamiltonian's definition (Eq. 2.1), we are left with

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \frac{\nabla_i^2 \psi}{\psi} + \frac{1}{2} \omega^2 \sum_{i=0}^{N-1} r_i^2 + \sum_{i<j} U_{m,n}(r_{ij}). \quad (2.9)$$

While the last terms are the mere potential energy terms, the first term involves the quantity $\psi^{-1} \nabla_i^2 \psi$, which can be rewritten noticing

$$\nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2 = \frac{\nabla_i^2 \psi}{\psi}. \quad (2.10)$$

This makes $\nabla_i \ln \psi$ a central quantity to evaluate. Expressing the position vector of the i -th particle as $\mathbf{r}_i = \sum_d \hat{\mathbf{x}}_d x_{id}$ and the vector pointing from the j -th particle to the i -th particle as $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j = r_{ij} \hat{\mathbf{r}}_{ij}$, we find

$$\begin{aligned} \nabla_i \ln \psi &= \sum_d \hat{\mathbf{x}}_d \frac{\partial}{\partial x_{id}} \ln \psi = \sum_d \hat{\mathbf{x}}_d \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right] \\ &= -\frac{\mathbf{r}_i}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{\mathbf{r}_{ij}}{r_{ij}^{\gamma+2}}. \end{aligned} \quad (2.11)$$

$$\begin{aligned}
 |\nabla_i \ln \psi|^2 &= \sum_d \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right]^2 = \\
 &= -\frac{r_i^2}{\alpha} - \frac{\gamma\beta^\gamma}{2\alpha} \sum_{j \neq i} \frac{r_i^2 - \mathbf{r}_i \cdot \mathbf{r}_j}{r_{ij}^{\gamma+2}} + \frac{\gamma^2\beta^{2\gamma}}{4} \sum_{j \neq i} \frac{1}{r_{ij}^{2\gamma+2}}
 \end{aligned} \tag{2.12}$$

$$\begin{aligned}
 \nabla_i^2 \ln \psi &= \sum_d \frac{\partial}{\partial x_{id}} \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right] \\
 &= -\frac{D}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{D-2-\gamma}{r_{ij}^{\gamma+2}}
 \end{aligned} \tag{2.13}$$

Finally, we conclude the analytical expression for the local energy is given by the following equation:

$$\begin{aligned}
 E_L(\mathbf{R}, \alpha, \beta, \gamma) &= -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \left[-\frac{D+r_i}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{D-2-\gamma-\alpha^{-1}r_i^2+\alpha^{-1}\mathbf{r}_i \cdot \mathbf{r}_j}{r_{ij}^{\gamma+2}} + \right. \\
 &\quad \left. + \frac{\gamma^2\beta^{2\gamma}}{4} \sum_{j \neq i} \frac{1}{r_{ij}^{2\gamma+2}} \right] + \frac{1}{2}\omega^2 R^2 + \sum_{i < j} 4\epsilon_{m,n} \left[\left(\frac{\sigma_{m,n}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{m,n}}{r_{ij}} \right)^6 \right]
 \end{aligned} \tag{2.14}$$

2.2.2 Cusp condition

While an approximate prediction for the optimal α parameter can be obtained easily by looking at the harmonic term (the ground state of a quantum harmonic oscillator has $\sqrt{\alpha} = \sqrt{\hbar/(m\omega)}$ as its characteristic oscillation length), in order to give an estimate of the other two parameters one may study the behavior of the local energy function we have just derived explicitly in Eq. 2.14.

For simplicity, first we consider a system composed of $N = 2$ particles. Looking at Eq. 2.14 and taking the limit $r_{1,2} \rightarrow 0$, we need to impose the following equality in order to suppress the higher-order energy divergences that make Eq. 2.14 far from uniform in space.

$$-\frac{\hbar^2\gamma^2\beta^{2\gamma}}{4mr_{1,2}^{2\gamma+2}} + 4\epsilon\tilde{\alpha} \left(\frac{\sigma}{r_{1,2}} \right)^{12} = 0 \tag{2.15}$$

This further implies

$$2\gamma + 2 = 12 \implies \gamma = 5$$

and, adhering to our convention $\frac{\hbar^2}{2m} = \frac{\epsilon\sigma^2}{10}$, we can also find a prediction for β :

$$\frac{\hbar^2}{2m} \left(\frac{25\beta^{10}}{2} \right) = \frac{\epsilon\sigma^2}{10} (40\tilde{\alpha}\sigma^{10}) \implies \beta = \left(\frac{16}{5}\tilde{\alpha} \right)^{1/10} \sigma. \tag{2.16}$$

For the symmetric case ($\tilde{\alpha} = 1$), this analytical condition predicts an optimal value of $\beta \approx 1.1227\sigma$.

It is important to point out that in the $r_{1,2} \rightarrow 0$ limit, the $1/r^{12}$ repulsive core overwhelmingly dominates the interaction. The Jastrow factor is specifically tailored to exactly cancel this highest-order singularity. Although subleading divergences—such as the $1/r^6$ attractive potential term or the $1/r^7$ contribution arising from the Laplacian—remain technically uncanceled, they pose no analytical or numerical threat. The optimized Jastrow factor $e^{-\beta/r_{1,2}^5}$ drives the trial wavefunction to zero exponentially fast as particles overlap, heavily suppressing any residual lower-order polynomial singularities in the evaluation of expectation values.

Moreover, it is worth noting that this result, derived for $N = 2$, remains strictly valid for any generic N . When any two particles i and j approach each other ($r_{ij} \rightarrow 0$), their pairwise interaction completely dominates the local energy. The presence of the other $N - 2$ particles yields only finite contributions that are negligible in this limit, making the two-body cusp condition universal for the Jastrow parameters regardless of the total number of particles in the droplet.

2.2.3 Dealing with different flavors

The trial wavefunction we have used up until here may be a good choice if $\tilde{\alpha} \approx 1$. However, if one is interested in different $\tilde{\alpha}$ values, it may be best to introduce a slightly different wavefunction:

$$\psi(\mathbf{R}; \alpha) = \prod_i e^{-\frac{r_i^2}{2\alpha}} \prod_{i < j} e^{-\frac{1}{2} \left(\frac{\beta_{m,n}}{r_{ij}} \right)^5} = \exp \left(-\frac{1}{2\alpha} \sum_i r_i^2 - \frac{1}{2} \sum_{i < j} \left(\frac{\beta_{m,n}}{r_{ij}} \right)^5 \right), \quad (2.17)$$

where m is the flavor of the i -th particle, n is the flavor of the j -th particle, $\beta_{AA} = \beta_{BB}$ and of course $\beta_{A,B} = \beta_{B,A}$. The math needed to utilize this wavefunction follows naturally from the previous one. Note that γ is no longer a variational parameter for the reasons discussed in Subsection 2.2.2; in fact, it is set to the constant value of $\gamma = 5$.

Other more technical aspects concerning dealing with different flavors will be discussed in Subsection 3.2.1.

2.3 Alternative methods to evaluate local kinetic energies

Throughout Subsection 2.2.1 we chose to express the kinetic term of the local energy utilizing both terms of the LHS of Eq. 2.10. An alternative option to obtain an estimate of the system's energy is noticing that the kinetic part of the Monte Carlo integral in Eq. 1.7,

$$\langle K_L \rangle = -\frac{\hbar^2}{2m} \frac{\int d\mathbf{R} |\psi|^2 \sum_{i=0}^{N-1} \frac{\nabla_i^2 \psi}{\psi}}{\int d\mathbf{R} |\psi|^2} = -\frac{\hbar^2}{2m} \frac{\int d\mathbf{R} |\psi|^2 \sum_{i=0}^{N-1} \nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2}{\int d\mathbf{R} |\psi|^2},$$

can be rewritten exploiting the following consequences of Green's Theorem.

$$\frac{\int d\mathbf{R} |\psi|^2 \nabla_i^2 \ln \psi}{\int d\mathbf{R} |\psi|^2} = \frac{\int d\mathbf{R} |\psi|^2 \nabla_i \cdot (\nabla_i \ln \psi)}{\int d\mathbf{R} |\psi|^2} = \text{B.T.} - 2 \frac{\int d\mathbf{R} |\psi|^2 |\nabla_i \ln \psi|^2}{\int d\mathbf{R} |\psi|^2},$$

where the boundary term (B.T.) rigorously vanishes because the trial wavefunction decays exponentially at spatial infinity due to the harmonic confinement. Hence we are left with

an equality between expectation values:

$$\langle \nabla_i^2 \ln \psi \rangle = -2 \langle |\nabla_i \ln \psi|^2 \rangle. \quad (2.18)$$

This allows us to find the expectation value of the local kinetic energy (and thereby of the local energy as well) averaging over the sampled values of even just one of the two quantities $\sum_{i=0}^{N-1} \nabla_i^2 \ln \psi$ and $\sum_{i=0}^{N-1} |\nabla_i \ln \psi|^2$.

$$\langle K_L \rangle = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \langle \nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2 \rangle \quad (2.19)$$

$$\langle K_L \rangle = -\frac{\hbar^2}{4m} \sum_{i=0}^{N-1} \langle \nabla_i^2 \ln \psi \rangle \quad (2.20)$$

$$\langle K_L \rangle = \frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \langle |\nabla_i \ln \psi|^2 \rangle \quad (2.21)$$

Expressions 2.19, 2.20 and 2.21 are all valid in expectation. Nonetheless, their statistical variances and computational costs differ notably: evaluating Eq. 2.21 requires computing only the first-order derivatives of the wavefunction, halving the analytical operations required per step, albeit potentially at the cost of a higher variance. It is thus recommendable to empirically test all three estimators to identify the most efficient trade-off.

2.4 Spatial Probability Densities

Finally, we aim to build an intuition regarding the spatial distribution of the system. More specifically, we want to answer the following questions:

1. *What is the probability density of finding **any** particle at a given point in space $\mathbf{r}$?*
2. *Observing the system from each particle's perspective, what is the probability density of finding **any other** particle at a given distance r ?*

2.4.1 One-Body Density

To answer the first question, we calculate the marginal probability density of a single particle by integrating the full N -body probability distribution over the position vectors of the remaining $N - 1$ particles.

Because the particles of a given flavor are indistinguishable bosons, the squared trial wavefunction $|\psi(\mathbf{R})|^2$ is symmetric under the exchange of any two particles of the same species. Furthermore, since our system is completely symmetric with respect to the two flavors ($N_A = N_B$, equal masses, and $U_{AA} = U_{BB}$), the spatial distribution of species A is physically identical to that of species B .

Consequently, computing the marginal probability density for a single generic particle and multiplying the result by the total number of particles N yields the total one-body density of the mixture, properly normalized to N :

$$\rho(\mathbf{r}) = N \frac{\int d\mathbf{r}_1 \dots d\mathbf{r}_{N-1} |\psi(\mathbf{r}, \mathbf{r}_1, \dots, \mathbf{r}_{N-1})|^2}{\int d\mathbf{R} |\psi(\mathbf{R})|^2}. \quad (2.22)$$

Since we are dealing with a many-body system, evaluating 2.22 requires a Monte Carlo approach. Conveniently, the same Metropolis algorithm devised for the VMC naturally distributes the N particles according to the target probability distribution $\propto |\psi(\mathbf{R})|^2$. This allows us to easily evaluate $\rho(\mathbf{r})$ by constructing a histogram.

For simplicity, we choose a radial histogram, meaning we divide the space into M concentric spherical shells of radii $\mathbf{r}_0, \mathbf{r}_1, \dots, \mathbf{r}_{M-1}$. At each Monte Carlo step, we measure the radial distance of every particle and increment the corresponding histogram bin. Working in D dimensions, the i -th bin counts the occurrences of particles whose positions $\mathbf{r}_j$ satisfy $\mathbf{r}_i \leq |\mathbf{r}_j| < \mathbf{r}_{i+1}$. To extract the probability $P(r)$ of finding a particle at distance r from the origin, we divide the final bin counts by the total number of Monte Carlo steps and by the bin width $\Delta \mathbf{r} = \mathbf{r}_{i+1} - \mathbf{r}_i$. Furthermore, we can find the local volumetric density $\rho(r)$, spherically averaged over the radius, by dividing the raw bin counts by the total Monte Carlo steps and by the exact geometric volume of the hyperspherical shell associated with that bin.

This same approach can be extended to concurrently obtain the one-body densities relative to each individual flavor group as well. Since our system is symmetric under a complete flavor exchange (i.e., all A 's swapping places with all B 's), as previously mentioned, we expect the one-body density of species A to perfectly overlap that of species B , with each contributing exactly half of the total $\rho(\mathbf{r})$.

On a last note, in order to provide a practical estimate of the statistical uncertainty for these distributions, we model the accumulation of particles in each bin as a Poisson process. Under this approximation, the standard error associated with a raw bin count C_i is given by $\sqrt{C_i}$. This absolute uncertainty is then scaled by the exact same geometric and Monte Carlo normalization factors used to compute $P(r)$ and $\rho(r)$, yielding an approximate error bar for our one-body density profiles. While the Poisson approximation provides a computationally inexpensive baseline for the statistical uncertainty, it inherently assumes independent sampling events. Because our configurations are generated via a Markov Chain, consecutive particle positions are autocorrelated, meaning the true statistical error is likely larger. A mathematically rigorous estimation would require applying the blocking algorithm to each spatial bin individually; however, storing the full histogram evolution over all Monte Carlo steps imposes a severe memory overhead and a better estimate of these uncertainties is beyond the scope of this project.

2.4.2 Pair Correlation Function

The second question is addressed by studying the two-body radial distribution function, known in the literature as the Pair Correlation Function, $g(r)$. While the one-body density describes the overall macroscopic shape of the droplet relative to the center of the harmonic trap, the pair correlation function reveals the internal microscopic structure of the many-body system by measuring the spatial correlation between any two particles.

Theoretically, $g(r)$ represents the conditional probability density of finding a particle at a distance r from another reference particle. In a quantum many-body system, this observable describes exactly how the presence of one atom locally modifies the spatial distribution of the surrounding ones due to their mutual interactions.

Given the binary nature of our mixture, this function naturally splits into two distinct physical quantities of interest:

- **Intra-species correlation ($g_{\text{alike}}(r)$):** describing the spatial correlation strictly between particles of the same species (i.e., A - A and B - B pairs);

- **Inter-species correlation** ($g_{\text{unlike}}(r)$): describing the spatial correlation between particles of different species (A - B pairs).

Comparing the theoretical profiles of $g_{\text{alike}}(r)$ and $g_{\text{unlike}}(r)$ provides direct quantitative evidence of the mixing properties of the quantum fluid. Deviations between these two distributions as the cross-interaction parameter $\tilde{\alpha}$ varies can highlight fundamental thermodynamic behaviors, such as fluid miscibility, spatial segregation, or the onset of phase separation between the two components.

Chapter 3

Implementation

3.1 The Code’s Objective and Structure

Our aim for this thesis is to study the neutral atom mixture system, whose background was laid out throughout Chapter 2. To accomplish this, we extended the VMC program for quantum mechanical systems, which had been previously designed and discussed in preliminary Projects [1] and [2] earlier this year¹. The Object-Oriented C++ program is meant to be efficient and modular, capable of being easily upgraded and re-utilized to study new physical scenarios or gain further insights into the ones already implemented. The output results can be graphically analyzed separately by complementary Python scripts.

The full source code is available at <https://github.com/DavideCacopardi/VMC-bachelor-thesis>. A standard `README.md` file is included and provides all necessary instructions for the user. Nevertheless, a brief overview of the program’s architecture is provided below to facilitate the comprehension of the simulation workflow and highlight key computational strategies.

3.1.1 Summary of the Main Routines

The execution of the program is subdivided into four routines:

1. *Optimization*: Starting from an initial parameter guess, the selected automatic optimization algorithm attempts to find the parameter vector which minimizes the set loss function $\mathcal{L} = E + w \text{Var}(E)$.
2. *FinalMC & Blocking*: A high-precision Monte Carlo energy estimation is performed² and the results are post-processed using the *Blocking* resampling technique.
3. *Spatial Distribution*: A One-Body Density histogram and a Pair Correlation histogram are output; additionally, the spatial trajectories of the particles during the Metropolis evolution are logged².

¹Note that Project [2] dealt with the use of neural networks as the system’s wavefunction. Here we shall not delve into those techniques; they are, however, still available as options in the present version of the code.

²Traditionally the optimal parameters generated by the *Optimization* step are used, but this can be overridden to allow for a manual parameter choice.

4. *Mesh Grid*: A linearly equally-spaced parameter grid is generated and a Monte Carlo energy estimation is run at each node. This is primarily aimed at gaining an intuitive understanding of the hyper-surface determined by the energy function $E(\alpha)$.

These routines can be selectively enabled via command-line interface (CLI) toggles, and will be executed in the aforementioned sequential order. All tunable parameters—encompassing both physical variables and technical settings—can be configured in the `config.json` file.

3.1.2 The Idea behind the Implemented Classes

The **System** class represents the complete quantum system. It acts as a bridge between the relative Hamiltonian, WaveFunction, Particle and Solver objects. It coordinates the Monte Carlo processes and returns the requested **Sampler** objects.

The **Hamiltonian** classes define the chosen physical environment interface and compute the Local Energy accordingly.

The **WaveFunction** classes define the chosen trial wavefunction and its optimizable parameters; analytical overrides for the relevant derivatives can be implemented here.

The **Particle** class defines the attributes (such as position and *flavor*) of each of the particles composing the physical system.

The **Solver** classes define the Metropolis algorithm employed to construct the Monte Carlo Markov Chain; more generally, they provide the step function called by the **System**'s methods.

The **Sampler** classes define what is to be sampled at each Monte Carlo step and handle the statistical math; as previously mentioned, they are output by the **System**'s methods as they retain all relevant simulation results, other than optionally logging them to file.

The **MCEngine** class wraps the Monte Carlo methods of the **System** class and handles the multi-threading execution.

The **VMCOptimizer** classes implement optimization algorithms manually or wrap imported ones from public-access libraries.

Other more technical classes and source files are well documented in the GitHub Repository.

3.2 Implementation Choices

In this section we explain the most notable algorithmic implementations from a computational standpoint.

3.2.1 Dealing with Different Flavors and Automating the Solvers

The `setupRandomUniformInitialState` function initializes the particles of the system. Traditionally, this solely implies generating random coordinates from a user-defined uniform distribution and optionally enforcing additional constraints such as setting all particles at a safe distance from each other (if dealing with strongly repulsive potentials). In our case we also need to assign a different *flavor* to each particle. We chose to assign flavor *A* to all evenly indexed particles and flavor *B* to the remaining half. One might alternatively assign a random flavor to each particle, relaxing the $N_A = N_B = N/2$ assumption; however, this

is better suited for a larger number of particles and lies beyond the scope of this thesis.

Furthermore, since our system’s observables depend on the particles’ flavors, special care must be taken regarding the Monte Carlo evolution of the particles. While bosonic symmetry is naturally respected by the standard Metropolis Markov Chain, isolated MCMC samplers can easily get trapped in metastable regions of the configuration space, leading to highly correlated samples and erratic energy estimates [10]. To ensure proper ergodicity and species mixing, we find it necessary to forcefully introduce a *particle swap* approximately and arbitrarily every ~ 10 *step* proposals of the Markov Chain, to ensure proper ergodicity and species mixing in the configuration space. This is implemented as a 1/10 stochastic flavor swap, utilizing the same acceptance criterion employed in Brute Force Metropolis (Subsection 1.3.2) because the flavor-swap proposal is perfectly symmetric.

Finally, we implemented a smart `stepLength/timeStep` auto-tuning system. It is a reasonable rule of thumb to try to set the `stepLength` parameter of a Brute Force Metropolis to a value yielding an acceptance ratio of approximately $\sim 50\%$. Likewise, it is advisable to set the `timeStep` parameter of an Importance Sampling Metropolis to target an acceptance ratio of roughly $\sim 90\%$. These optimal step values depend considerably on the physical system and configurations being analyzed; traditionally, this forces the user to manually restart the VMC code several times to find a suitable parameter choice. To circumvent this, we implemented an automatic `step`-parameter tuning, acting every ~ 100 equilibration Metropolis steps, applying a simple 10% increase or decrease based on the acceptance ratio measured during the preceding ~ 100 steps. Note that this tuning must occur strictly during the equilibration phase, because changing the `step` parameter in the middle of a Metropolis sampling process compromises the detailed balance, thereby possibly altering the accuracy of the sampling itself; this bias is not notably impactful for the simple purpose of the equilibration steps.

3.2.2 Optimization of Parameters

Three main optimization algorithms are implemented: BFGS, Adam and Nelder-Mead. While BFGS and Nelder-Mead implementations are wrappers around the NLOpt open-source library [5], the Adam algorithm was implemented manually³.

Automated Number of Monte Carlo Cycles Increase

Since precise Monte Carlo runs can take a considerable amount of time, we implemented the capability to start our optimization algorithms with a relatively low number of Monte Carlo steps in order to quickly reach the neighborhood of the minimum. Only subsequently are we interested in refining the search with more Monte Carlo cycles, especially when the energy well is not steep. All three algorithms feature an automated increase of Monte Carlo cycles. Every `c_wait` = 10 optimization steps, the following signal-to-noise ratio condition is checked.

$$|\mathcal{L}_i - \mathcal{L}_{i-1}| < 4\sigma_E \quad (3.1)$$

If it evaluates to `true`, the number of Monte Carlo cycles is increased by 20% and the check is repeated. This increase can be triggered for a maximum of `c_max_improvement_tries` = 5

³Adam is also the optimization algorithm of choice to train the neural networks. For that specific purpose, the PyTorch library [8] and environment were utilized; however, as previously stated, this falls outside the scope of the present thesis.

consecutive times. Please note that this automated increase system is a ‘black-box’ to the optimization algorithm. That is, the optimization algorithm only sees the most recent evaluation of $\mathcal{L}_i$ in order not to compromise the validity of more than one optimization step. The loss function $\mathcal{L} = E + w \text{Var}(E)$ —the driver of the optimization algorithm—depends on good energy and gradient estimates, and possibly even on the evaluated energy variance and its gradient in turn, too. Growing the number of Monte Carlo cycles throughout the process can abruptly reduce the stochastic noise, making the newly evaluated $\mathcal{L}_i$ appear deceptively lower than $\mathcal{L}_{i-1}$, potentially stalling the algorithm if not handled properly. Finally, it is worth noting that the RHS of Ineq. 3.1 is a very rough estimate of the uncertainty on $\mathcal{L}_i$. A rigorous evaluation would require not only sampling the error on $\text{Var}(E)$, but also running a resampling technique (such as the *Blocking* algorithm) to account for the inherent covariance of the Markov Chain. We opt to bypass such high-precision calculations here, as this automated step-increase machinery merely aims to accelerate a process that would otherwise require manual trial-and-error.

Stopping Criteria

Both `NLOpt` wrappers use `set_xtol_rel` as their primary stopping criterion (usually set to $\sim 10^{-5}$); this means that the optimization process terminates when the length of the parameter step $(\Delta\alpha)_i = \alpha_i - \alpha_{i-1}$ is such that $(\Delta\alpha)_i / \alpha_i < \text{xtol_rel}$. Additional fallback `NLOpt` stopping criteria are enforced via `set_maxeval(400)` and `set_maxtime(10800)`.

Conversely, the Adam optimizer always executes a user-defined number `Adam_nSteps` of optimization steps.

On top of all these, the SIGINT ‘external interrupt’ listener is redirected to gracefully terminate the optimization process when it is still underway, enabling the user to skip the remaining optimization steps and proceed to the next routines without losing progress.

3.2.3 Learning Rate Reduction on Plateau

Although Adam is an adaptive learning rate scheme in itself, we implemented a supplementary loss-based learning-rate adjuster, in order to force a more precise convergence when a plateau is reached.

If the condition

$$\mathcal{L}_i < \left(\min_{j \leq i} \mathcal{L}_j \right) - \text{Adam_min_improvement}$$

evaluates to `false` for `Adam_max_patience` consecutive optimization steps, the learning rate is divided by 2 and the `min_improvement` parameter by 10.

3.2.4 Multi-threading & Blocking

As anticipated in Subsections 3.1.1 and 3.1.2, the code is designed to execute parallel Monte Carlo samplings across a user-defined number of threads (`nThreads`). The resulting `Sampler` objects are then averaged together into a single comprehensive output `Sampler`. Running multiple independent MCMC samplers in parallel and combining their samples is a primary strategy not only to speed up computation, but to directly reduce the final statistical variance of the estimators [10]

As for the Blocking resampling step, our code utilizes Jonsson’s C++ implementation, available at <https://github.com/computative/block>. There, the confidence level is set to 5%.

This algorithm requires the total dataset size to be a power of 2, as explained in Subsection 1.5.1. The number of Monte Carlo cycles for the *FinalMC* and *Mesh Grid* runs can indeed be specified by the user as any power of 2. However, when utilizing multiple threads, the actual number of Monte Carlo steps executed by each individual thread is a $1/\mathbf{nThreads}$ fraction of the total. Since the Blocking algorithm must individually resample each contiguous batch of data produced by each thread, the number of Monte Carlo cycles run by each thread must also be a power of 2. Consequently, if one wishes to obtain a Blocking uncertainty estimation, **nThreads** must be chosen as a power of 2 itself.

Chapter 4

Execution, Data and Results

4.1 Parameter Choice and Analytical Predictions

To benchmark our Variational Monte Carlo code and investigate the physical properties of the neutral atom mixture, we define a consistent set of algorithmic configurations and physical parameters, deriving key analytical reference values to guide the analysis of our numerical results.

Unless otherwise specified, all simulations employ Importance Sampling Metropolis dynamics stabilized by the Umrigar drift (Subsection 1.3.4) with interpolation parameter $a = 1$, alongside the stochastic particle-flavor swap routine (discussed in Subsection 3.2.1). We study three-dimensional systems ($D = 3$) composed of $N = 2, 4, 6, 8$ particles of identical unit mass $m = 1$, governed by Hamiltonians configured as follows.

4.1.1 Lennard-Jones Potential

For the Lennard-Jones potential, we adopt standard unit parameters: $\sigma = 1$, $\epsilon = 1$. To satisfy our energy-scale convention $\frac{\hbar^2}{2m} = \frac{\epsilon\sigma^2}{10}$, the kinetic diffusion constant is fixed to $D = \frac{\hbar^2}{2m} = 0.1$. To explore different interaction regimes, we analyze three values of the cross-interaction factor: $\tilde{\alpha} = 1.00, 0.40, 0.05$.

Setting the derivative of a Lennard-Jones interaction to zero, one can find its minimum:

$$r_{ij}^{(\min)} = 2^{1/6}\sigma = 2^{1/6} \approx 1.122, \quad (4.1)$$

$$\begin{aligned} U_{AA}(r_{ij}^{(\min)}) &= -\epsilon = -1, \\ U_{AB}(r_{ij}^{(\min)}) &= -\tilde{\alpha}\epsilon = -\tilde{\alpha}. \end{aligned} \quad (4.2)$$

4.1.2 Harmonic Potential

As for the harmonic trap potential, we select two distinct angular frequencies ω , reported in Table 4.1.

The ground state of a 3D quantum harmonic oscillator has energy $E_0 = \frac{3}{2}\hbar\omega$ and a characteristic Gaussian width $\sqrt{\alpha} = \sqrt{\frac{\hbar}{m\omega}}$. The energy of a 3D system of N particles is thus $E_N = \frac{3}{2}N\hbar\omega$. The α and E_0 values corresponding to the chosen ω values are also defined and listed in Table 4.1.

ω	α	E_0
$\omega_1 := 0.111803$	4	0.074 999 732 4
$\omega_2 := 0.027951$	16	0.018 750 100 8

TABLE 4.1: Quantum harmonic potential predictions for parameter α and ground state energies E_0 .

4.2 Non-Interacting System

In order to test the efficacy of our code, we first disable the interaction potential and benchmark the computed energies against the analytical solutions for independent harmonic oscillators.

The optimization algorithms require the initial guess to be set within the same strictly convex region as the sought-after minimum and may easily fail due to the great stochastic noise imposed by non-optimal parameter neighborhoods. Therefore, from now on, we choose to run the implemented Mesh Grid procedure as a way to double check the optimization results; in fact, the equally-spaced manual grid-search provides a safe rough estimate of the optimal set of parameters. If the automated optimization algorithm fails to converge to the same minimum found by the Mesh Grid method (within a margin of error), then one can repeat the optimization process employing a different algorithm or tailoring the optimization parameters to the observed behavior.

N	ω	$E^{(\text{expected})} = NE_0$	optimizer	w	$\alpha^{(0)}$	α	E
2	ω_1	0.149 999 464 9	Nelder-Mead	0.05	4	4.0023	0.149 999 50(1)
2	ω_1	0.149 999 464 9	Nelder-Mead	0.05	4	3.9976	0.149 999 48(2)
2	ω_1	0.149 999 464 9	BFGS	0.05	4	4.0000	0.149 999 464 9(1)
2	ω_1	0.149 999 464 9	Mesh Grid	0.00	–	4.0(4)	0.149 999 464 0(4)
4	ω_1	0.299 998 929 7	BFGS	0.05	4	4.0000	0.299 998 929 8(2)
4	ω_1	0.299 998 929 7	Mesh Grid	0.00	–	4.0(4)	0.299 998 928 9(8)
6	ω_1	0.449 998 394 6	BFGS	0.05	4	4.0000	0.449 998 394 4(3)
6	ω_1	0.449 998 394 6	Mesh Grid	0.00	–	4.0(4)	0.449 998 395(1)
8	ω_1	0.599 997 859 4	BFGS	0.05	4	4.0000	0.599 997 859 4(4)
8	ω_1	0.599 997 859 4	Mesh Grid	0.00	–	4.0(4)	0.599 997 860(1)
2	ω_2	0.037 500 201 6	BFGS	0.05	12	16.000	0.037 500 201 63(7)
2	ω_2	0.037 500 201 6	Mesh Grid	0.00	–	16.0(4)	0.037 500 201 5(2)
4	ω_2	0.075 000 403 2	BFGS	0.05	16	16.000	0.075 000 403 1(1)
4	ω_2	0.075 000 403 2	Mesh Grid	0.00	–	16.0(4)	0.075 000 403 3(3)
6	ω_2	0.112 500 604 9	BFGS	0.05	16	16.000	0.112 500 605 0(1)
6	ω_2	0.112 500 604 9	Mesh Grid	0.00	–	16.0(5)	0.112 500 604 7(5)
8	ω_2	0.150 000 806 5	BFGS	0.05	16	16.000	0.150 000 806 4(2)
8	ω_2	0.150 000 806 5	Mesh Grid	0.00	–	16.0(5)	0.150 000 806 5(7)

TABLE 4.2: VMC results for non-interacting harmonic $V = \frac{1}{2}\omega^2 R^2$ systems. Different optimization methods were tested. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Both optimization algorithms used $\mathbf{x.tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.

The results are shown in Table 4.2 and perfectly agree with the analytical predictions¹. Notice how the BFGS optimization converges to the expected result with remarkable accuracy, even when initialized far from the optimal value (e.g., $\alpha^{(0)} = 12$). Nelder-Mead, on the other hand, exhibits minor residual deviations on the order of $\sim 10^{-3}$, primarily arising from simplex stagnation in the presence of stochastic noise.

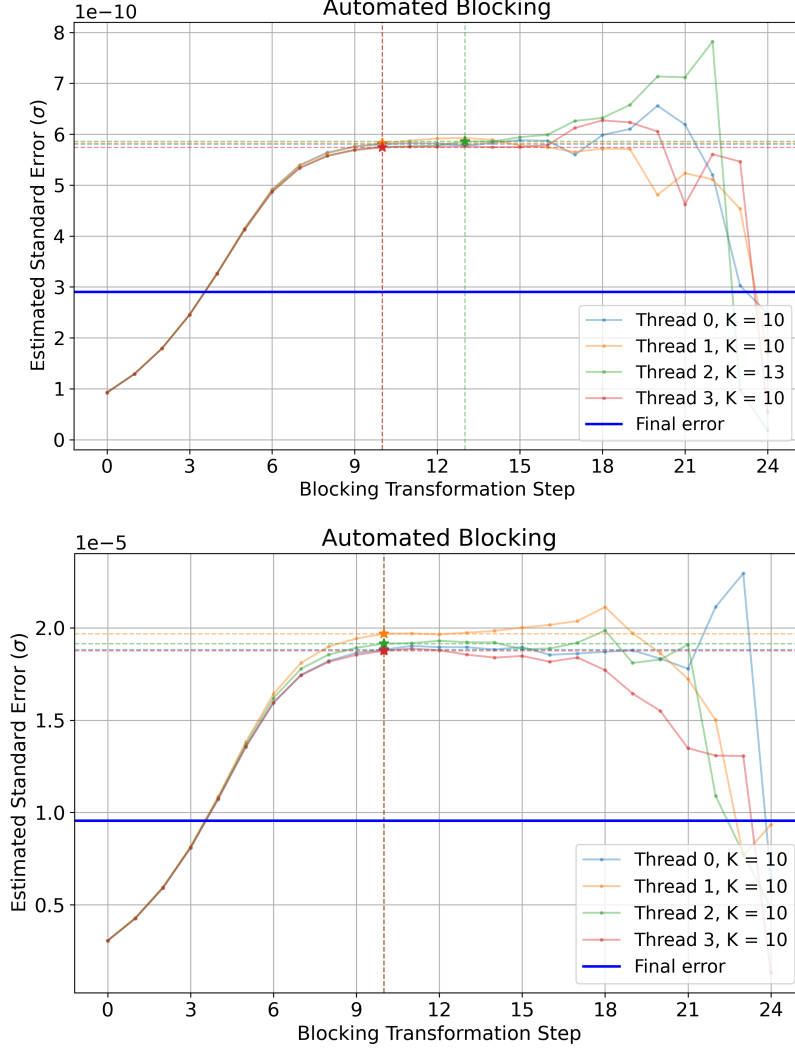


FIGURE 4.1: Estimated standard error vs number of blocking transformations via the automated blocking algorithm (Subsection 3.2.4). The first plot analyzes the final MC (2^{27} MC cycles) run for the $N = 6$, ω_1 non-interacting system (Table 4.2); the second plot refers to a final MC run for the same system but manually setting a non-optimal $\alpha = 4.5$, resulting in an energy estimate of 0.45311 ± 0.00001 as opposed to the naive pre-blocking result 0.453107 ± 0.000002 .

¹Further analysis on non-interacting systems utilizing a previous version of this program were conducted in Project [1].

Figure 4.1 shows how Marius Jonsson’s *Blocking* algorithm successfully detects the plateau of the apparent variance trend in each data-set. On the other hand, the uncertainty associated to the optimal parameters obtained from automated optimization algorithm is harder to estimate: several factors should be taken into account, including the energy gradients evaluated at convergence and the stochastic noise which affects them. Because of this hereon we choose to list the found optimal parameters up to 5 significant digits (since we usually set $\text{x.tol} = 10^{-5}$), but more conservative and realistic estimates would probably stop at the 3rd significant digit, assessing the qualitative flatness of the energy around the minimum and the stochastic errors during optimization.

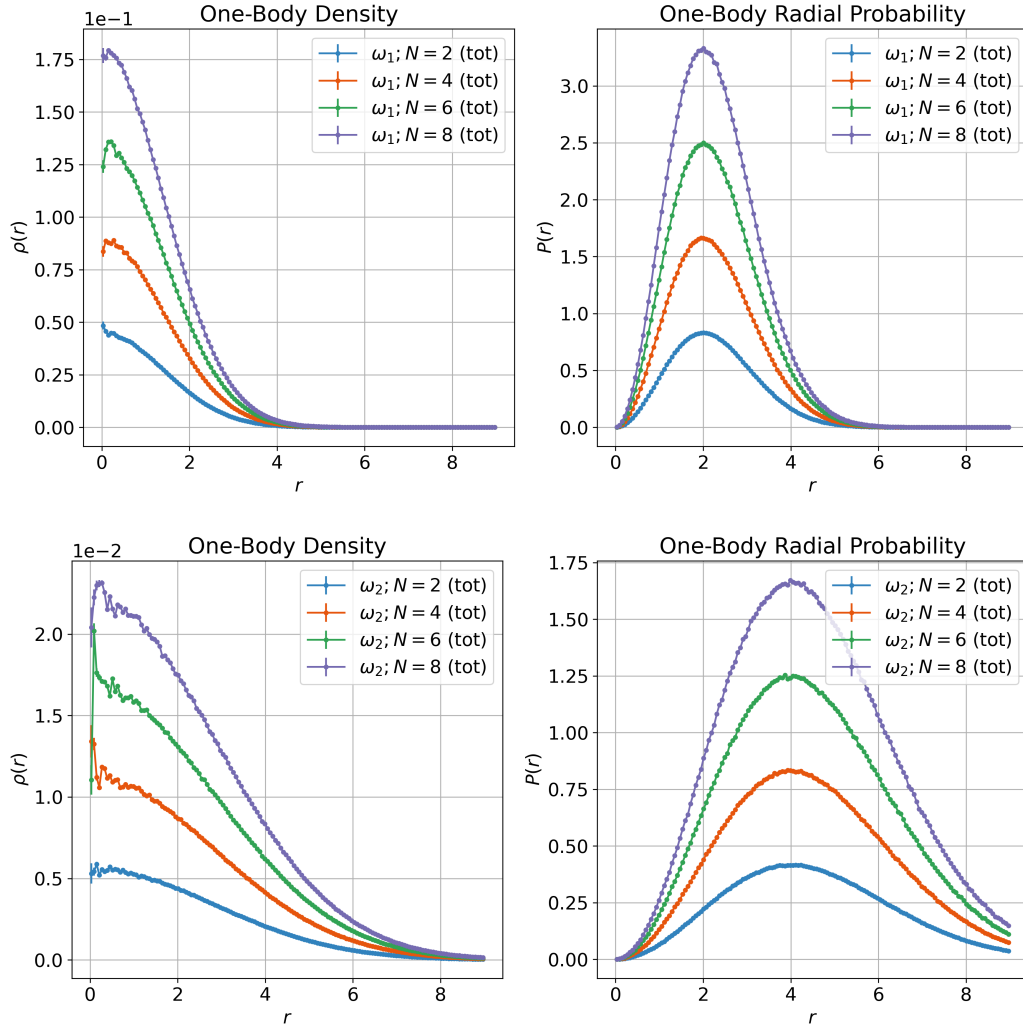


FIGURE 4.2: Radial one-body densities ($\sim 10^7$ MC steps) of the non-interacting systems. As expected, these non-normalized plots scale with the number of particles and perfectly match the gaussian widths $\sqrt{\alpha} = 2, 4$.

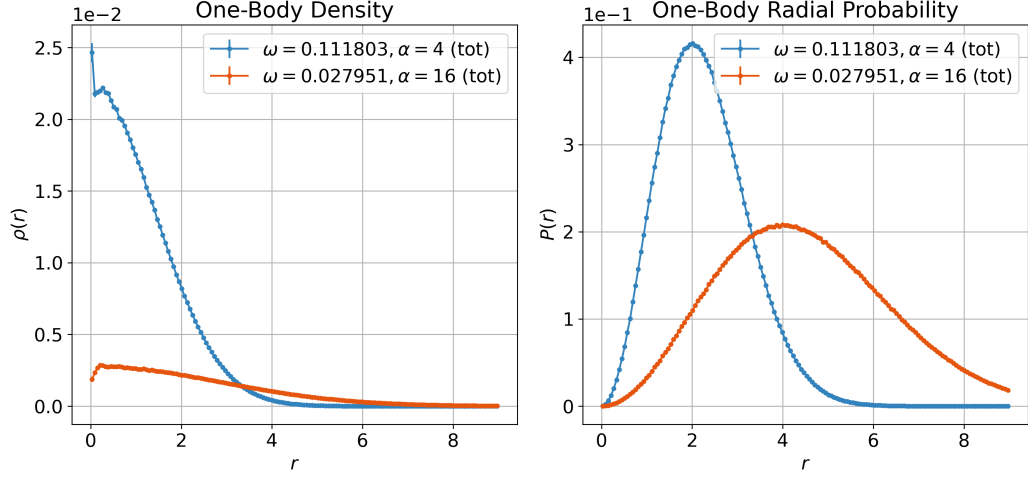


FIGURE 4.3: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 4$ non-interacting systems. The plot is normalized by the number of particles. The stronger the confinement ω , the tighter the distribution.

Finally, comparative tests on the local kinetic energy estimators confirmed that Eq. 2.20 yields a lower statistical variance than Eq. 2.21, while the complete laplacian formulation of Eq. 2.19 consistently exhibits the minimal variance overall. An explicit comparison is later shown in Figure 4.14 for an interacting system. Consequently, Eq. 2.19 is adopted as the primary kinetic estimator throughout all subsequent simulations.

4.3 Interacting Systems

4.3.1 Strong confinement

Next, we investigate the fully interacting Lennard-Jones mixture under harmonic confinement.

We first consider the strong confinement regime $\omega = \omega_1$. The results are shown in Table 4.3 and the most relevant ones are visualized in Figure 4.4; some one-body density plots and pair correlation plots are presented in Figures 4.8 to 4.16.

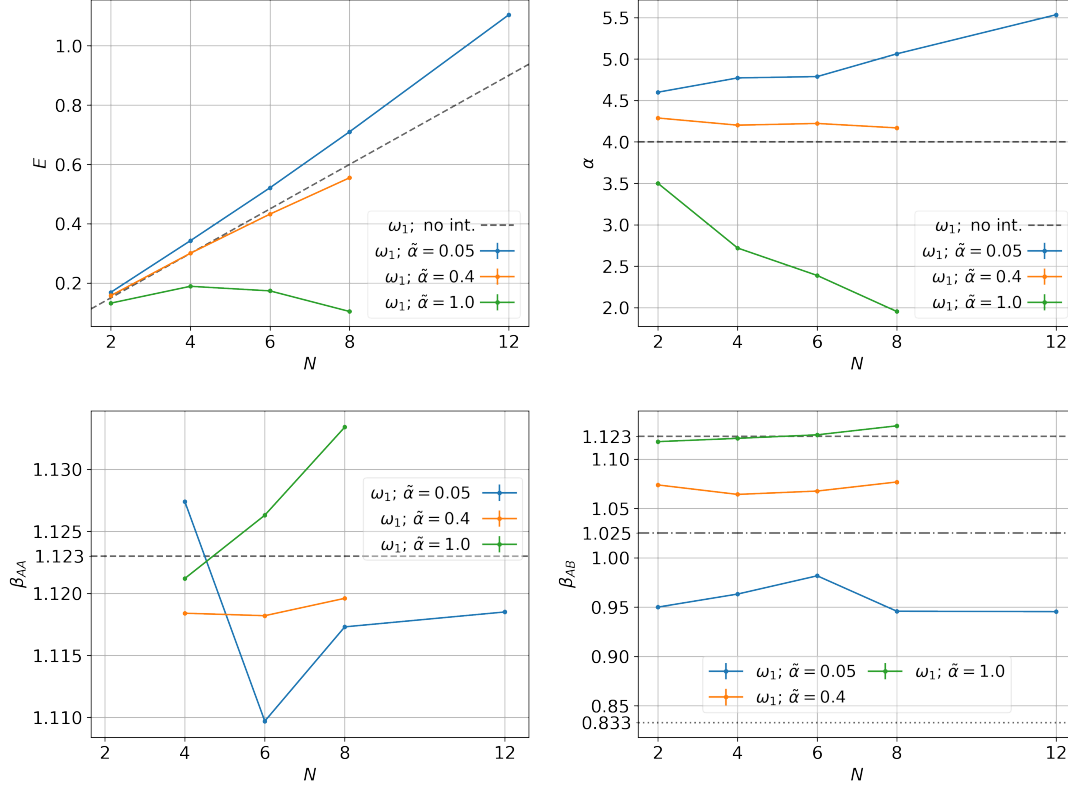


FIGURE 4.4: Best energy, α and β values obtained for the interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement.

All measured energies exhibit a low relative error ($\lesssim 0.01\%$). According to the Vanishing-Variance Principle (Subsection 1.2.2), this proves that the chosen trial wavefunction provides a robust variational ansatz for the interacting ground state, up to a seemingly satisfying degree of approximation for our subsequent observations.

After a few failed attempts at employing gradient-based optimizers, we had to resort to Nelder-Mead in spite of its well-known drawbacks. The gradient of the variance had often overwhelmed that of the energy due to noise and generally tended to lead both the α and β 's parameters to higher values in order to disperse the bosons and thereby prevent sampling points close to the singularities, even when it would actually be energetically favorable to do so. Only at the end did we manage to find better results by lowering the weight w parameter and starting the gradient-based optimization closer to the previously identified

TABLE 4.3: VMC results for interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement. Different optimization methods were tested. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05, 0.05, 0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Adam ran 120 steps with 10^7 MC cycles and an initial learning rate of 0.03. Nelder-Mead and BFGS used $\mathbf{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.

N	$\tilde{\alpha}$	optimizer	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
2	1.00	Nelder-Mead	0.05	4	—	1	3.5043	—	1.1169	0.132 306(8)
2	1.00	BFGS	0.05	3.5	—	1.1	3.4999	—	1.1177	0.132 305(8)
2	1.00	Mesh Grid	0.00	—	—	—	3.6(2)	—	1.08(7)	0.132 24(2)
4	1.00	Nelder-Mead	0.05	2.3	1	1	2.6703	1.1227	1.1240	0.189 19(3)
4	1.00	Nelder-Mead	0.05	2.3	1	1	2.7195	1.1212	1.1210	0.189 03(3)
4	1.00	Adam	0.05	2.3	1	1	2.7472	1.1193	1.1180	0.189 04(3)
4	1.00	Mesh Grid	0.00	—	—	—	2.6(3)	1.12(4)	1.12(4)	0.1883(3)
6	1.00	Nelder-Mead	0.05	2.3	1	1	2.3881	1.1263	1.1246	0.173 59(7)
6	1.00	Mesh Grid	0.00	—	—	—	2.2(2)	1.15(7)	1.15(7)	0.1830(3)
8	1.00	Nelder-Mead	0.05	2.3	1	1	1.9532	1.1334	1.1337	0.1041(1)
8	1.00	Mesh Grid	0.00	—	—	—	2.0(2)	1.15(7)	1.15(7)	0.1121(6)
2	0.40	Nelder-Mead	0.05	4.3	—	1	4.2884	—	1.0738	0.157 485(8)
2	0.40	Mesh Grid	0.00	—	—	—	4.2(3)	—	1.08(7)	0.157 50(4)
4	0.40	Nelder-Mead	0.05	4.3	1	1	4.2020	1.1184	1.0642	0.300 43(3)
4	0.40	Mesh Grid	0.00	—	—	—	4.2(3)	1.12(4)	1.08(4)	0.2997(1)
6	0.40	Nelder-Mead	0.05	4.3	1	1	4.2228	1.1182	1.0676	0.432 46(5)
6	0.40	Mesh Grid	0.00	—	—	—	3.9(3)	1.08(7)	1.08(7)	0.4337(5)
8	0.40	Nelder-Mead	0.05	4.3	1	1	4.1691	1.1196	1.0768	0.5549(1)
8	0.40	Mesh Grid	0.00	—	—	—	3.9(3)	1.15(7)	1.08(7)	0.5599(3)
2	0.05	Nelder-Mead	0.05	4.5	—	1	4.4407	—	0.9423	0.168 98(2)
2	0.05	Nelder-Mead	0.05	4.6	—	1	4.6000	—	0.9500	0.168 71(2)
2	0.05	Mesh Grid	0.00	—	—	—	4.6(2)	—	1.01(7)	0.168 63(9)
4	0.05	Nelder-Mead	0.05	4.4	1	1	4.3110	1.1169	0.9346	0.345 22(6)
4	0.05	Nelder-Mead	0.05	4.6	1	1	4.5084	1.1169	0.9410	0.343 60(6)
4	0.05	Nelder-Mead	0.05	4.7 [†]	1.1	1	4.7722	1.1274	0.9632	0.342 78(6)
4	0.05	Mesh Grid	0.00	—	—	—	4.8(4)	1.12(4)	1.00(4)	0.3425(4)
6	0.05	Nelder-Mead	0.05	4.6	1	1	4.4306	1.1188	0.9401	0.5259(1)
6	0.05	Nelder-Mead	0.05	4.8 [†]	1.1	1	5.1093	1.1261	0.9334	0.5259(1)
6	0.05	Adam	0.0005	4.8	1.1	1	4.7884	1.1097	0.9817	0.5213(1)
6	0.05	Mesh Grid	0.00	—	—	—	4.8(4)	1.16(12)	1.04(12)	0.5304(4)
8	0.05	Nelder-Mead	0.05	4.6	1	1	4.8579	1.1174	0.9481	0.7094(2)
8	0.05	Nelder-Mead	0.05	4.8 [†]	1.1	1	5.0631	1.1173	0.9458	0.7101(2)
8	0.05	Mesh Grid	0.00	—	—	—	4.8(4)	1.16(12)	1.04(12)	0.7226(6)
12*	0.05	Nelder-Mead	0.05	4.8	1.11	0.83	5.5925	1.1202	0.9372	1.1109(3)
12*	0.05	Nelder-Mead	0.05	5.3	1.1	0.94	5.3442	1.1185	0.9454	1.1044(3)
12*	0.05	Mesh Grid	0.00	—	—	—	5.0(5)	1.112**	0.95(7)	1.098(2)

* Bonus run.

[†] Nelder-Mead ran with $\pm [0.5, 0.05, 0.05]$ as initial state setting.

** $\beta_{AA} = 1.112$ was held constant throughout the grid-search (i.e., the parameter was not optimized).

optimal neighborhood. Further investigation might be needed in order to refine the gradient evaluation, employ a better loss function or simply tune the gradient-based optimizers more finely.

A first analysis of the Jastrow parameters, β_{AA} and β_{AB} , highlights how they roughly follow the trend predicted in Subsection 2.2.2 to cancel the $\sim 1/r_{ij}^{12}$ dependence from the local energy. Indeed, evaluating Eq. 2.16 with our chosen parameters predicts:

$$\beta(\tilde{\alpha} = 1.00) \approx 1.123; \quad \beta(\tilde{\alpha} = 0.40) \approx 1.025; \quad \beta(\tilde{\alpha} = 0.05) \approx 0.833;$$

and we have found:

$$\beta_{AB}(\tilde{\alpha} = 1.00) = 1.124(3); \quad \beta_{AB}(\tilde{\alpha} = 0.40) = 1.071(2); \quad \beta_{AB}(\tilde{\alpha} = 0.05) = 0.957(6).$$

The most striking aspect is the behavior of the α parameter and of the energy. Not only is α heavily influenced by the $\tilde{\alpha}$ asymmetry parameter, but it is also notably dependent on the number of particles N . An explanation for this phenomenon can be found by analyzing the 2-body problem and subsequently extending the reasoning to a larger number of particles. From Eq. 4.1, we know the minimum of the Lennard-Jones interaction is achieved once the pair of particles is found at a relative distance of $r_{ij} \approx 1.122$. This value is smaller than the Gaussian width $\sqrt{\alpha}$ imposed on all the particles by the harmonic confinement. Heuristically speaking, we can thus claim that the Lennard-Jones potential tends to draw particles closer together than the natural width of the trap, which favors a smaller spatial variance ($\alpha < 4$). In practice, we observe this holds true only for $\tilde{\alpha} = 1.00$, whereas the $\tilde{\alpha} = 0.40$ and $\tilde{\alpha} = 0.05$ systems exhibit the very opposite behavior. This is because we need to take into account the Zero Point Energy of the 2-particle non-interacting system and compare it to the depth of the Lennard-Jones potential well we found in Eq. 4.2. The 2-body Hamiltonian

$$H = -\frac{\hbar^2}{2m}\nabla_i^2 - \frac{\hbar^2}{2m}\nabla_j^2 + \frac{1}{2}m\omega^2(r_i^2 + r_j^2) + U_{ij}(r_{ij})$$

can indeed be rewritten in the following form:

$$H = -\frac{\hbar^2}{2(2m)}\nabla_{\text{CM}}^2 + \frac{1}{2}(2m)\omega^2 r_{\text{CM}}^2 - \frac{\hbar^2}{2(m/2)}\nabla_{ij}^2 + \frac{1}{2}(m/2)\omega^2 r_{ij}^2 + U_{ij}(r_{ij}).$$

From this last equation, it is evident that the relative motion of the two particles in 3D has a ZPE of $E_{ij} = \frac{3}{2}\hbar\omega = E_0$. Given our current choice $\omega = \omega_1$, this means that $\tilde{\alpha} \gtrsim E_0 \approx 0.075$ must hold for a pair of particles of different flavors to be bound together by the Lennard-Jones interaction; otherwise, they are repelled by the $\sim 1/r_{ij}^{12}$ potential wall. Likewise, all pairs of particles alike in flavor show an attractive behavior since $\epsilon = 1 \gg E_0 = 0.075$.

The next step, as anticipated earlier, is to address how the overall effect of the Lennard-Jones potential on the system varies as the number of particles grows. The number of interactions between particles of the same flavor is

$$N_{\text{alike}} = \frac{N}{2} \left(\frac{N}{2} - 1 \right) = \frac{N^2}{4} - \frac{N}{2}$$

and the number of those between particles of different flavors is

$$N_{\text{unlike}} = \left(\frac{N}{2} \right)^2 = \frac{N^2}{4}.$$

Additionally we know that the harmonic contribution to the energy grows linearly with N . If this were all, we could only successfully explain the data observed for the $\tilde{\alpha} = 1.00$ and $\tilde{\alpha} = 0.05$ cases.

In the former, the $N = 2$ system is composed of only one (A, B) attractive pair (because $\tilde{\alpha} > E_0$) and no 'alike' pairs. $\alpha \approx 3.5 < 4$ is found, hence the particles are more tightly bound than in the respective non-interacting system. Furthermore, as the number of particles N grows, the system is more and more tightly bound since the number of attractive interactions scales as $\sim N^2$, unlike the $\propto N$ harmonic confinement contribution.

In the latter $\tilde{\alpha} = 0.05$ case, the $N = 2$ system is composed only of an (A, B) repulsive pair (because $\tilde{\alpha} < E_0$) and no 'alike' pairs. $\alpha \approx 4.5 > 4$ is found, hence the particles are more dispersed than in the respective non-interacting system. Furthermore, as the number of particles N grows, the system seems to keep slowly enlarging. Naively assuming that each repulsive interaction exactly cancels one attractive interaction would model a $\propto (N_{\text{unlike}} - N_{\text{alike}}) = \frac{N}{2}$ linear growth, as opposed to the $\propto (N_{\text{unlike}} + N_{\text{alike}}) \sim \frac{N^2}{2}$ quadratic shrinking of the totally attractive $\tilde{\alpha} = 1.00$ case.

The outlier is the $\tilde{\alpha} = 0.40$ dataset: although the N_{unlike} pairs should contribute in a lightly attractive manner ($\tilde{\alpha} > E_0$) and the gaussian width α undeniably shrinks as N grows, all analyzed systems show $\alpha > 4$.

While this is confusing at first, the missing puzzle piece lies in the optimized values of the β_{AB} parameter, which systematically overestimate the analytical predictions in the $\tilde{\alpha} < 1$ cases. Eq. 2.16 predicts that as $\tilde{\alpha}$ reduces, β_{AB} shrinks too, in order to follow the more gentle $\sim 1/r_{ij}^{12}$ wall. In other words, particles of different flavors should be able to approach each other more closely. However, for all we have just discussed, once $\tilde{\alpha}$ is lower than the E_0 threshold, the depth of the U_{AB} potential well is not enough for the attractivity of the potential to be expressed. Consequently, even though the $\sim 1/r_{ij}^{12}$ barrier tightens around smaller r_{ij} , the two particles must keep at a longer distance (i.e., β_{AB} should increase). A variational 'tug-of-war' emerges, leading the optimizer to settle on a trade-off: the β_{AB} we ultimately measure.

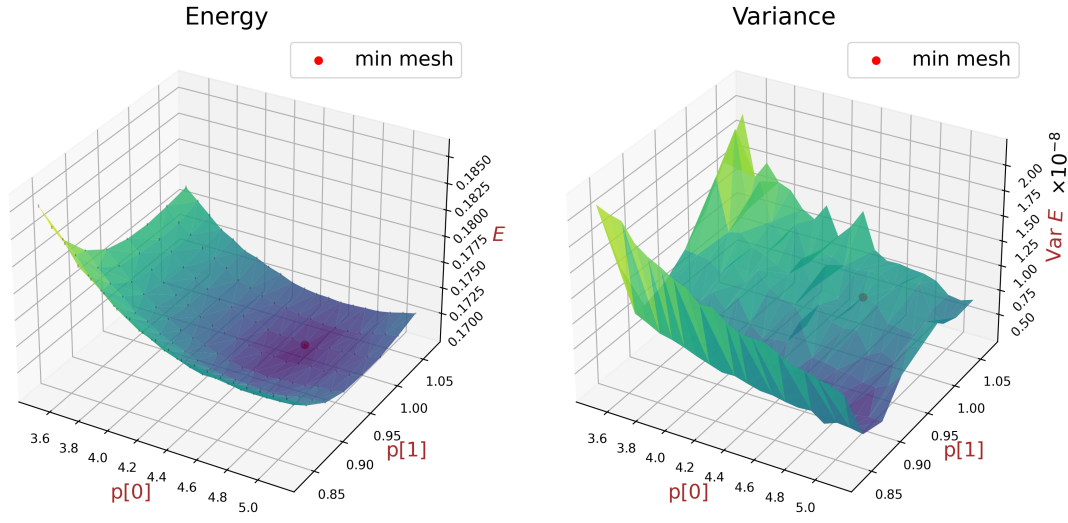


FIGURE 4.5: Mesh Grid run ($\sim 10^7$ MC steps) on an $N = 2$ system governed by a $\omega = \omega_1$ harmonic trap and a $\tilde{\alpha} = 0.05$ Lennard-Jones potential. $\mathbf{p}[0]$ is the α parameter; $\mathbf{p}[1]$ is the β_{AB} parameter. The minimum-energy point is highlighted in red both in the energy plot and in the variance plot. It is evident how the variance valley does not mimic the energy valley: the minimum variance is achieved by a lower β_{AB} and a generally higher α value than the optimal ones for the energy.

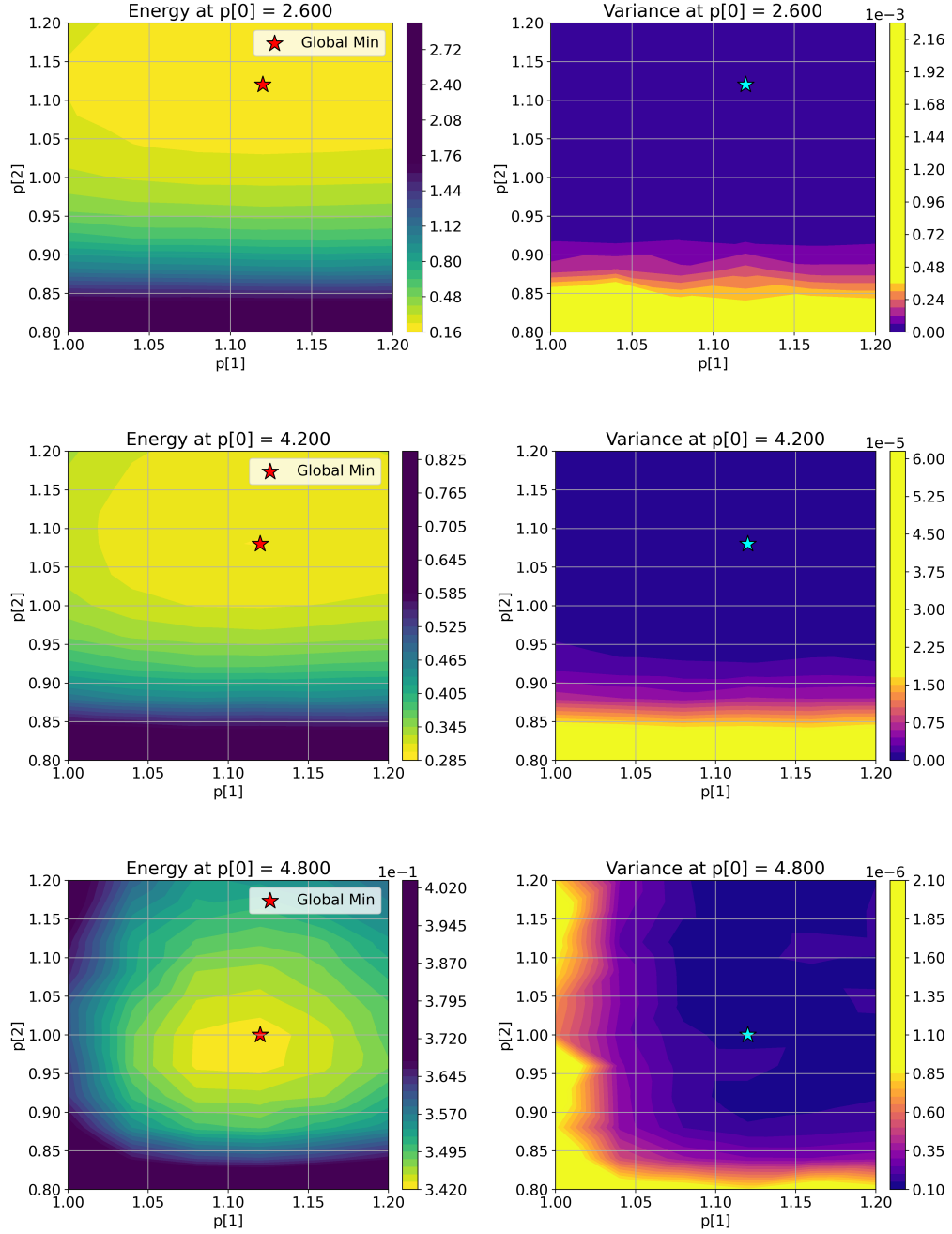


FIGURE 4.6: Comparison of Mesh Grid runs ($\sim 10^7$ MC steps) on the $N = 4$ systems governed by a $\omega = \omega_1$ harmonic trap. The top figures depict a Mesh Grid slice of the $\tilde{\alpha} = 1.00$ system; the middle figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.40$ system; the bottom figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.05$ system. $p[0]$ is the α parameter; $p[1]$ is the β_{AA} parameter; $p[2]$ is the β_{AB} parameter. The minimum-energy point is highlighted each energy plot and in the corresponding variance plot. It is evident how the variance valley does not perfectly mimic the energy valley.

This claim is further reinforced by comparing the variance trend against the energy with the help of the Mesh Grid visualization: the variance valley does not strictly correspond to the energy valley (see Figures 4.6 and 4.5). According to the Vanishing-Variance Principle (Subsection 1.2.2), this is a symptom of inexact representation of the true ground state wavefunction.

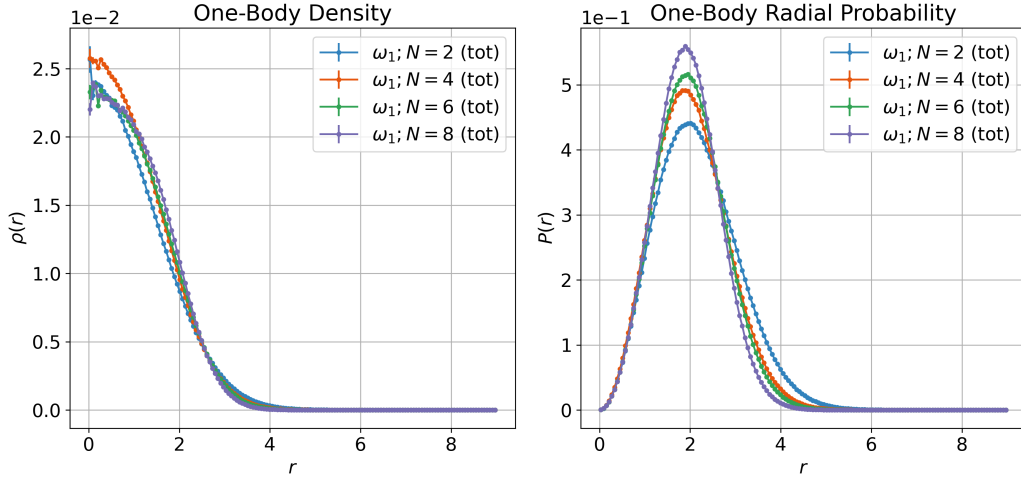


FIGURE 4.7: Radial one-body densities ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential.

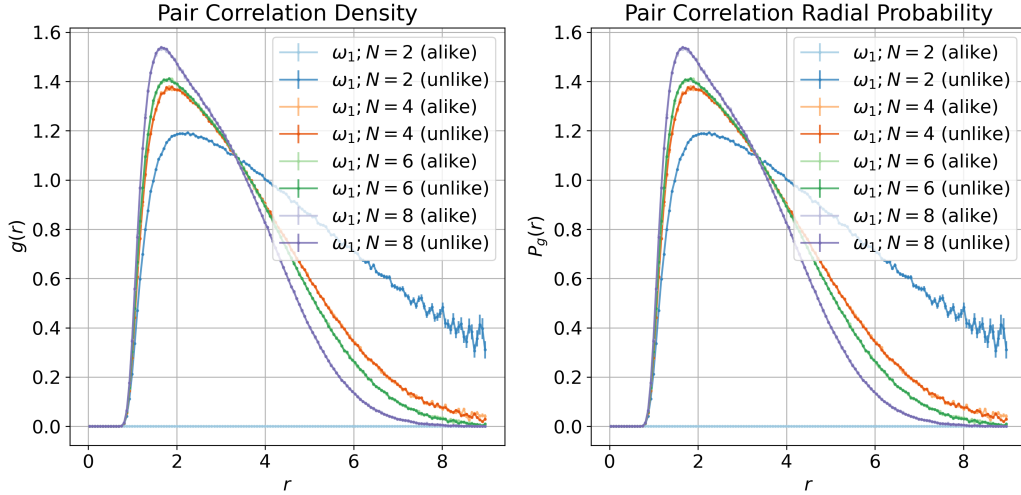


FIGURE 4.8: Radial pair correlation function ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. This is normalized by the pair correlation function observed in the corresponding non-interacting systems.

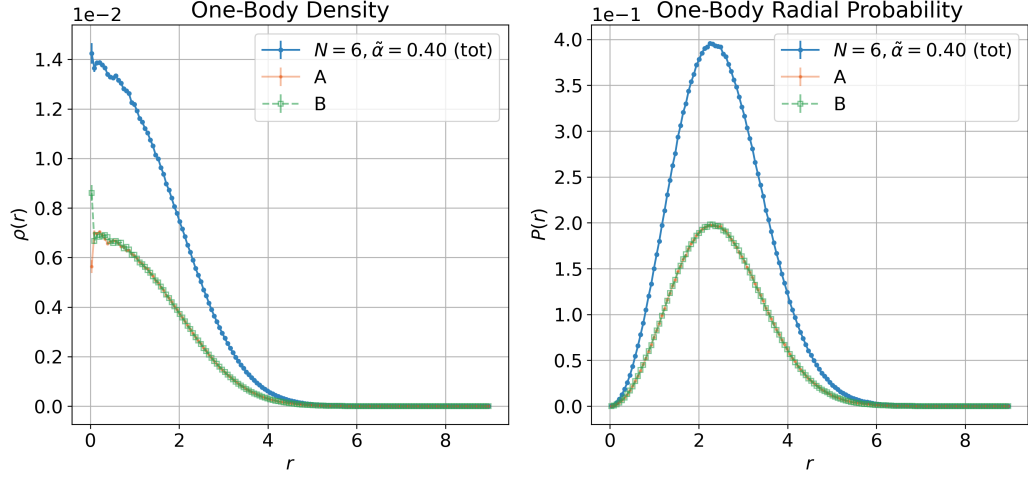


FIGURE 4.9: Radial one-body density ($\sim 10^7$ MC steps) of the $N = 6$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.40$ Lennard-Jones potential. OBDs of the A particles and B particles are also individually plotted. All plots are normalized by the same number of particles $N = 6$. It is evident that the OBD of species A perfectly overlaps that of species B , and each contributes exactly half of the total $\rho(r)$ as expected (Subsection 2.4.1).

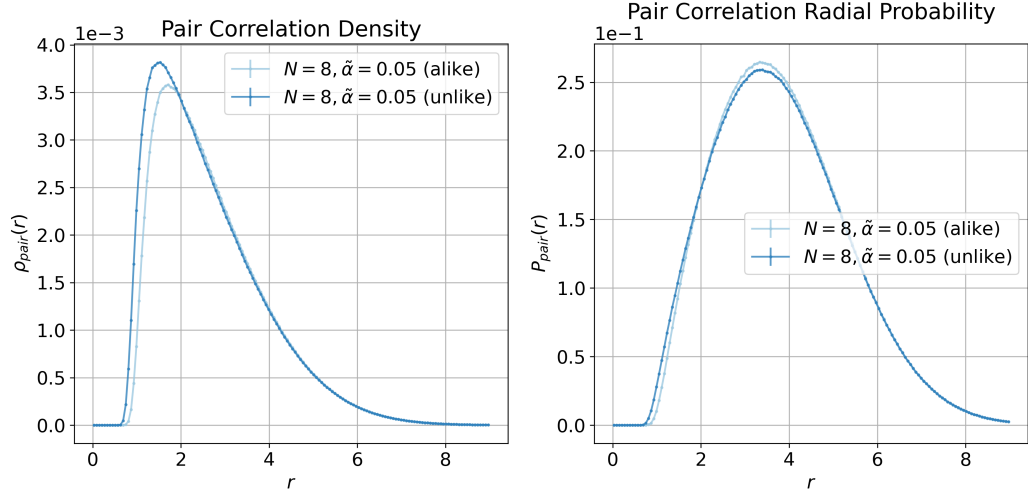


FIGURE 4.10: Radial pair correlation function ($\sim 10^7$ MC steps) of the $N = 8$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.05$ Lennard-Jones potential. OBDs of the A particles and B particles are also individually plotted. Each plot is normalized by the number of unique interaction pairs of its relative "type" (alike or unlike). It is evident how $\beta_{AB} < \beta_{AA}$ draws particles of different flavors together.

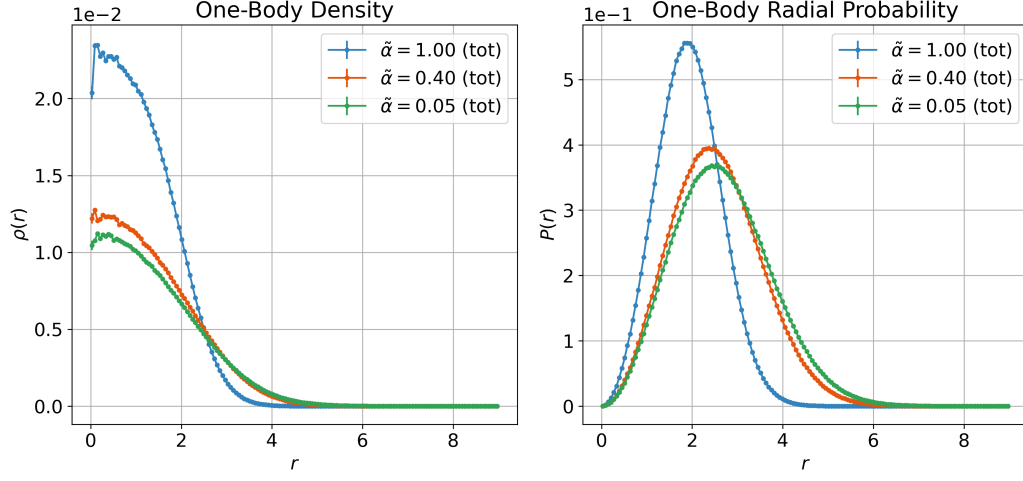


FIGURE 4.11: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap. The plot is normalized by the number of particles. It is evident how a stronger Lennard-Jones interaction between different species allows for an overall more compact system.

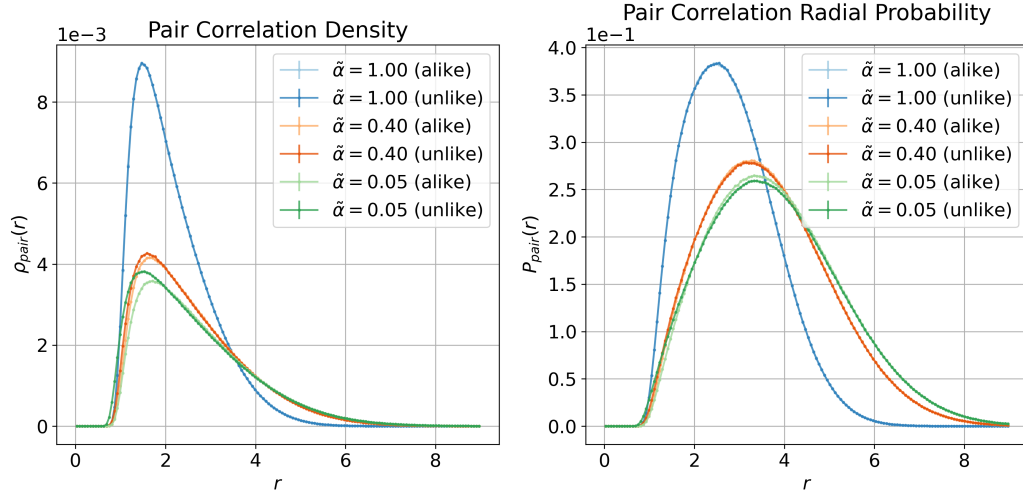


FIGURE 4.12: Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap. Each plot is normalized by the number of unique interaction pairs of its relative "type" (alike or unlike). It is evident how $\beta_{AB} < \beta_{AA}$ draws particles of different flavors together. The $\tilde{\alpha} = 0.05$ case is plotted individually in Figure 4.10.

4.3.2 Weak confinement

Now we proceed by running the VMC code setting $\omega = \omega_2 = 0.027951$.

The results shown in Table 4.4 and Figure 4.13 reinforce everything we have stated so far even more firmly.

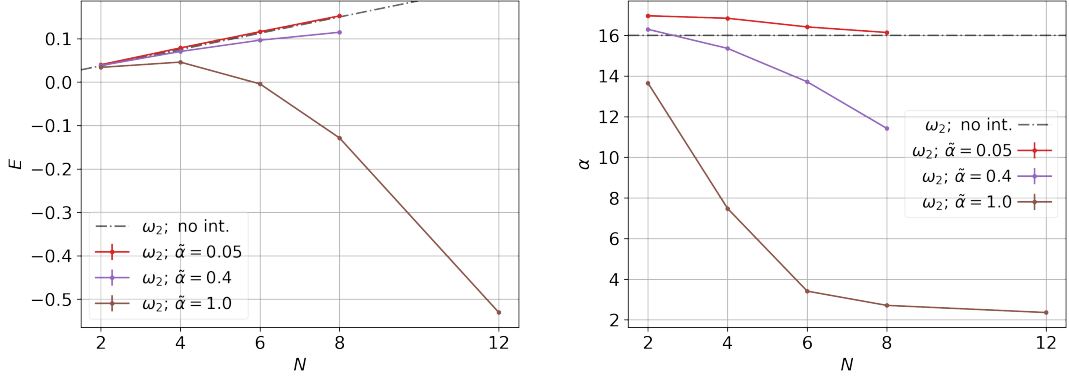


FIGURE 4.13: Best energy and alpha values obtained for the interacting systems inside a $\omega = \omega_2 = 0.027951$ harmonic confinement.

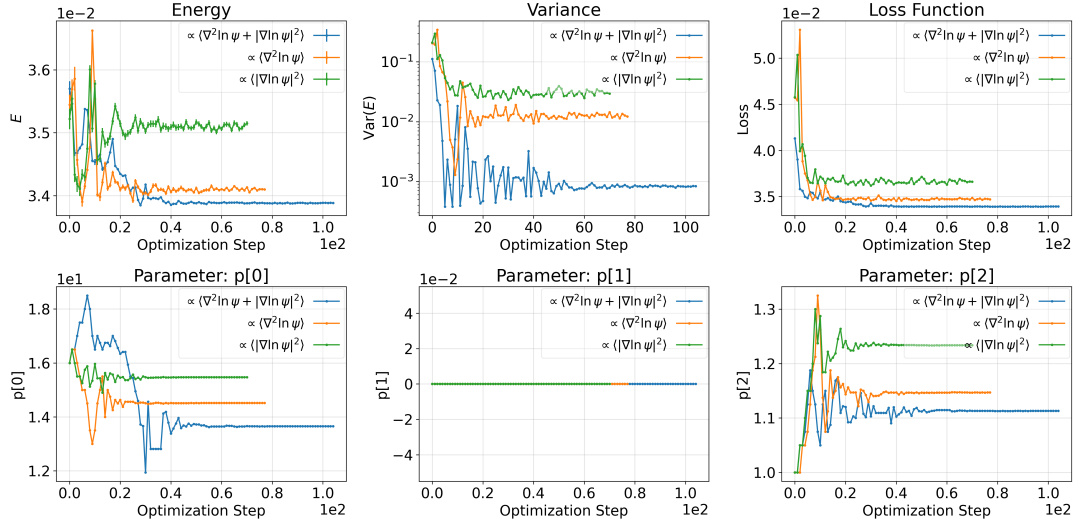


FIGURE 4.14: Comparison of Nelder-Mead optimization processes for the $N = 2$ interacting system inside a $\omega = \omega_2$ harmonic confinement. The blue optimization utilized Eq. 2.19 to evaluate the local kinetic energy; the orange one utilized Eq. 2.20; the green one utilized Eq. 2.21. Parameters $p[0]$, $p[1]$ and $p[2]$ correspond to α , β_{AA} and β_{AB} , respectively.

TABLE 4.4: VMC results for interacting systems inside a $\omega = \omega_2$ harmonic confinement. Different optimization methods were tested. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.50, 0.05, 0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Both optimization algorithms used $\mathbf{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.

N	$\tilde{\alpha}$	optimizer	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
2	1.00	Nelder-Mead	0.05	16	—	1	13.651	—	1.1130	0.033 880(4)
2	1.00	Nelder-Mead [†]	0.05	16	—	1	14.513	—	1.1469	0.034 10(1)
2	1.00	Nelder-Mead [‡]	0.05	16	—	1	15.471	—	1.2336	0.035 13(2)
2	1.00	Mesh Grid	0.00	—	—	—	13.7(2)	—	1.12(6)	0.033 89(1)
4	1.00	Nelder-Mead	0.05	16	1.1	1.1	7.4787	1.1127	1.1165	0.046 00(2)
4	1.00	Mesh Grid	0.00	—	—	—	7.5(5)	1.10(5)	1.10(5)	0.0458(1)
6	1.00	Nelder-Mead	0.05	16	1.1	1.1	3.5038	1.1241	1.1228	−0.004 02(2)
6	1.00	Nelder-Mead**	0.05	3.5038	1.1241	1.1228	3.4118	1.1236	1.1207	−0.004 16(6)
6	1.00	Mesh Grid	0.00	—	—	—	4(1)	1.1(1)	1.1(1)	−0.0022(4)
8	1.00	Nelder-Mead	0.05	16	1	1	2.7701	1.1296	1.1268	−0.1282(1)
8	1.00	Nelder-Mead**	0.05	2.7701	1.1296	1.1268	2.7110	1.1239	1.1261	−0.1287(1)
8	1.00	Mesh Grid	0.00	—	—	—	3(1)	1.1(1)	1.1(1)	−0.1211(9)
12*	1.00	Nelder-Mead	0.05	2	1.1	1.1	2.3599	1.1301	1.1314	−0.5304(2)
12*	1.00	Mesh Grid	0.00	—	—	—	2.5(5)	1.12(4)	1.12(4)	−0.5269(10)
2	0.40	Nelder-Mead	0.05	16	—	1	16.298	—	1.0666	0.038 021(2)
2	0.40	Mesh Grid	0.00	—	—	—	16.5(5)	—	1.06(6)	0.038 01(1)
4	0.40	Nelder-Mead	0.05	16	1.1	1.1	15.360	1.1156	1.0697	0.070 471(6)
4	0.40	Mesh Grid	0.00	—	—	—	15.0(5)	1.12(6)	1.06(6)	0.070 48(4)
6	0.40	Nelder-Mead	0.05	13	1.1	1	13.716	1.1108	1.0646	0.096 46(2)
6	0.40	Mesh Grid	0.00	—	—	—	13.0(5)	1.12(6)	1.06(6)	0.096 48(8)
8	0.40	Nelder-Mead	0.05	16	1	1	11.424	1.1106	1.0706	0.114 76(4)
8	0.40	Mesh Grid	0.00	—	—	—	12.0(5)	1.12(6)	1.06(6)	0.1154(1)
2	0.05	Nelder-Mead	0.05	17	—	1	16.966	—	0.9459	0.039 984(7)
2	0.05	Mesh Grid	0.00	—	—	—	17.5(5)	—	1.00(5)	0.0399(4)
4	0.05	Nelder-Mead	0.05	16	1	1	16.842	1.1051	0.9411	0.078 69(2)
4	0.05	Mesh Grid	0.00	—	—	—	17.0(5)	1.12(6)	1.00(5)	0.0784(2)
6	0.05	Nelder-Mead	0.05	16	1.1	1	16.417	1.1083	0.9875	0.116 01(5)
6	0.05	Mesh Grid	0.00	—	—	—	16.5(5)	1.12(6)	0.95(10)	0.1164(2)
8	0.05	Nelder-Mead	0.05	16	1.1	1	16.143	1.1192	0.9559	0.152 57(8)
8	0.05	Mesh Grid	0.00	—	—	—	16.5(5)	1.12(6)	0.95(10)	0.1525(3)

[†] Local kinetic energy evaluated with Eq. 2.20.

[‡] Local kinetic energy evaluated with Eq. 2.21.

* Bonus run.

** These runs were executed subsequently to those of Table 4.5 for the sake of fair comparison.

Because the harmonic trap is substantially broader ($\alpha = 16$), the kinetic zero-point energy barrier is significantly lowered. Consequently, all evaluated $\tilde{\alpha}$ cross-interaction strengths now strictly satisfy the heuristic binding condition ($\tilde{\alpha} > E_0$); hence, all observed regimes display a globally cohesive behavior: across all datasets, α decreases as N grows. Analyzing the Jastrow parameters, the variational ‘tug-of-war’ is once again visible. The weak cross-interactions ($\tilde{\alpha} = 0.05$ and $\tilde{\alpha} = 0.40$) still drive the optimizer to slightly inflate β_{AB} above the theoretical cusp predictions to artificially push the uncoupled species apart.

This regime ultimately unveils the most striking result of our study: for $\tilde{\alpha} = 1.00$, the system undergoes a crossover from a trapped quantum gas into a self-bound quantum liquid droplet. As the particle number increases, the Gaussian width collapses dramatically from $\alpha \approx 13.6$ down to $\alpha \approx 2.7$. At $N = 6$ and $N = 8$, the total ground-state energy inverts its ascending trend and crosses below zero ($E \approx -0.004$ and $E \approx -0.128$, respectively). Since both the expected values of the quantum kinetic energy and the external harmonic potential are strictly positive quantities, measuring an overall negative energy is mathematically sufficient to affirm that the cohesive Lennard-Jones attraction has overwhelmingly dominated the physics of the system, indicating that the system has structurally transitioned into a self-bound quantum liquid droplet. In this limit, the droplet’s spatial size is governed entirely by the internal Lennard-Jones balance rather than the external trap walls. The cluster forms a cohesive, self-bound liquid that would remain structurally stable even in the limit of vanishing confinement ($\omega \rightarrow 0$).

4.3.3 No confinement

We can benchmark this structural stability further by running the VMC program one last time: we set $\omega = 0$ —effectively removing the harmonic confinement—and initiate the optimization process from the optimal parameters found for the $\omega = \omega_2, \tilde{\alpha} = 1.00$ systems.

TABLE 4.5: Successful VMC results for interacting systems without any confinement. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.50, 0.05, 0.05]$ as initial state setting and $\times 10^7$ MC cycles. BFGS ran with 10^7 MC cycles. Both optimization algorithms used $\mathbf{x}_{\text{tol}} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.

N	$\tilde{\alpha}$	optimizer	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
6	1.00	Nelder-Mead	0.05	3.5038	1.1241	1.1228	3.7838	1.1220	1.1184	$-0.019\,28(6)$
6	1.00	Mesh Grid	0.00	—	—	—	3.7(7)	1.12(4)	1.12(4)	$-0.0189(2)$
8	1.00	Nelder-Mead	0.05	2.7701	1.1296	1.1268	2.8182	1.1249	1.1260	$-0.1459(1)$
8	1.00	Mesh Grid	0.00	—	—	—	3.0(3)	1.12(4)	1.12(4)	$-0.1465(4)$
12*	1.00	Nelder-Mead	0.05	2.3	1.12	1.12	2.5078	1.1336	1.1298	$-0.5544(2)$
12*	1.00	Mesh Grid	0.00	—	—	—	2.3(3)	1.12(4)	1.12(4)	$-0.551(1)$

* Bonus run.

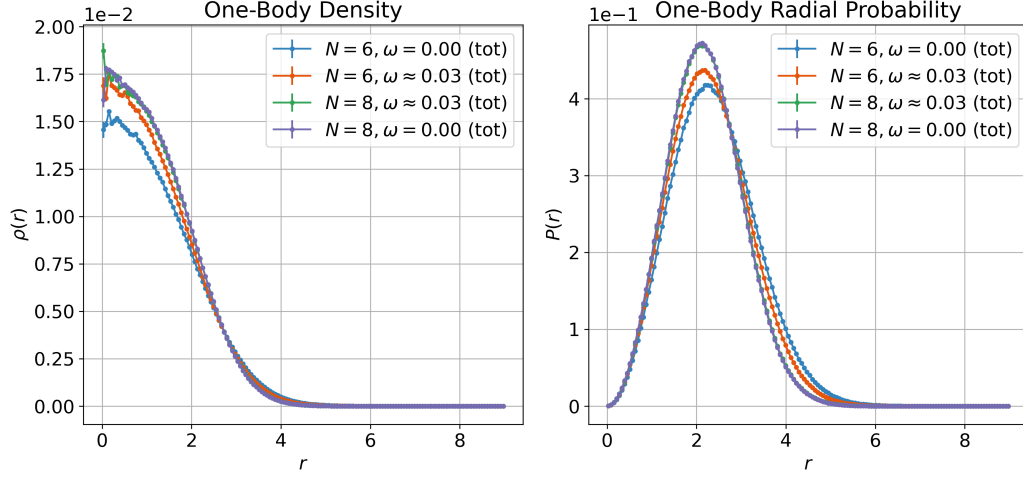


FIGURE 4.15: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. The plots are normalized by the number of particles.

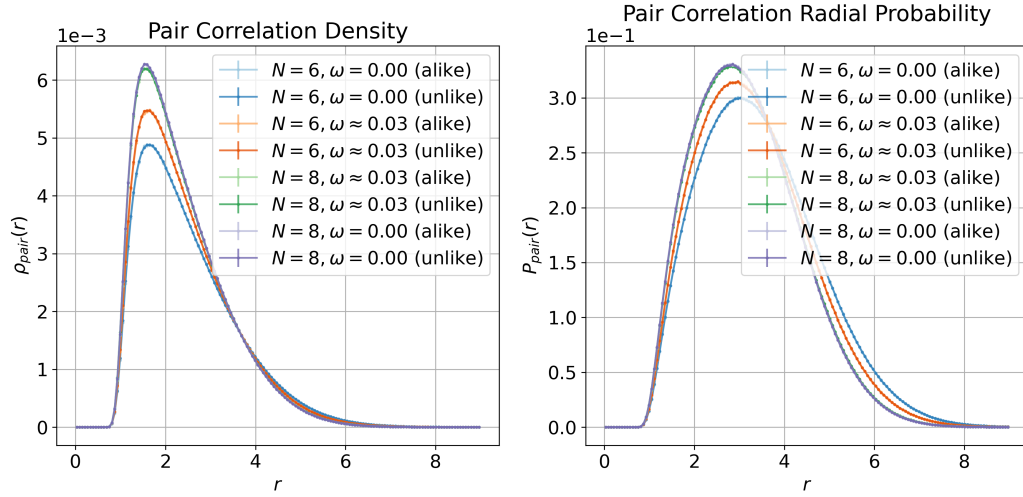


FIGURE 4.16: Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. Each plot is normalized by the number of unique interaction pairs of its relative "type" (alike or unlike).

The results for the systems which successfully converged are shown in Table 4.5; namely, $N = 6$ and $N = 8$ systems remained completely intact. Conversely, the $N = 2$ and $N = 4$ systems struggled to find a minimum and resorted to disgregating the bosonic cloud as much as possible².

This diverging behavior perfectly mirrors theoretical expectations for quantum droplets. As demonstrated by De Palo et al. [3], in order to have a self-bound stable liquid droplet in free space, the chemical potential μ must be strictly negative to prevent the system from evaporating. The chemical potential essentially dictates the energetic cost of adding or removing particles. For low particle numbers ($N = 2, 4$), the cohesive forces are insufficient to yield a negative chemical potential, causing the unconfined cloud to simply evaporate into the vacuum. As N increases to 6 and 8, the addition of particles actively lowers the system's overall energy (implying $\mu < 0$), securing the structural stability of the droplet even in the absence of a trap.

As predicted, for these successfully bound states, the optimal α parameters closely resemble the ones we found under weak confinement, proving that the previous systems did not struggle to prevail over the harmonic potential. Lastly, the total energies are lowered since the positive contributions due to the harmonic term have been cleanly eliminated.

²Virtually $\alpha \rightarrow \infty$, though the numerical upper-bound was set to $\alpha = 50$. Furthermore, we ran the same test for the $N = 8, \tilde{\alpha} = 0.40$ system and, likewise, it did not converge to any finite α .

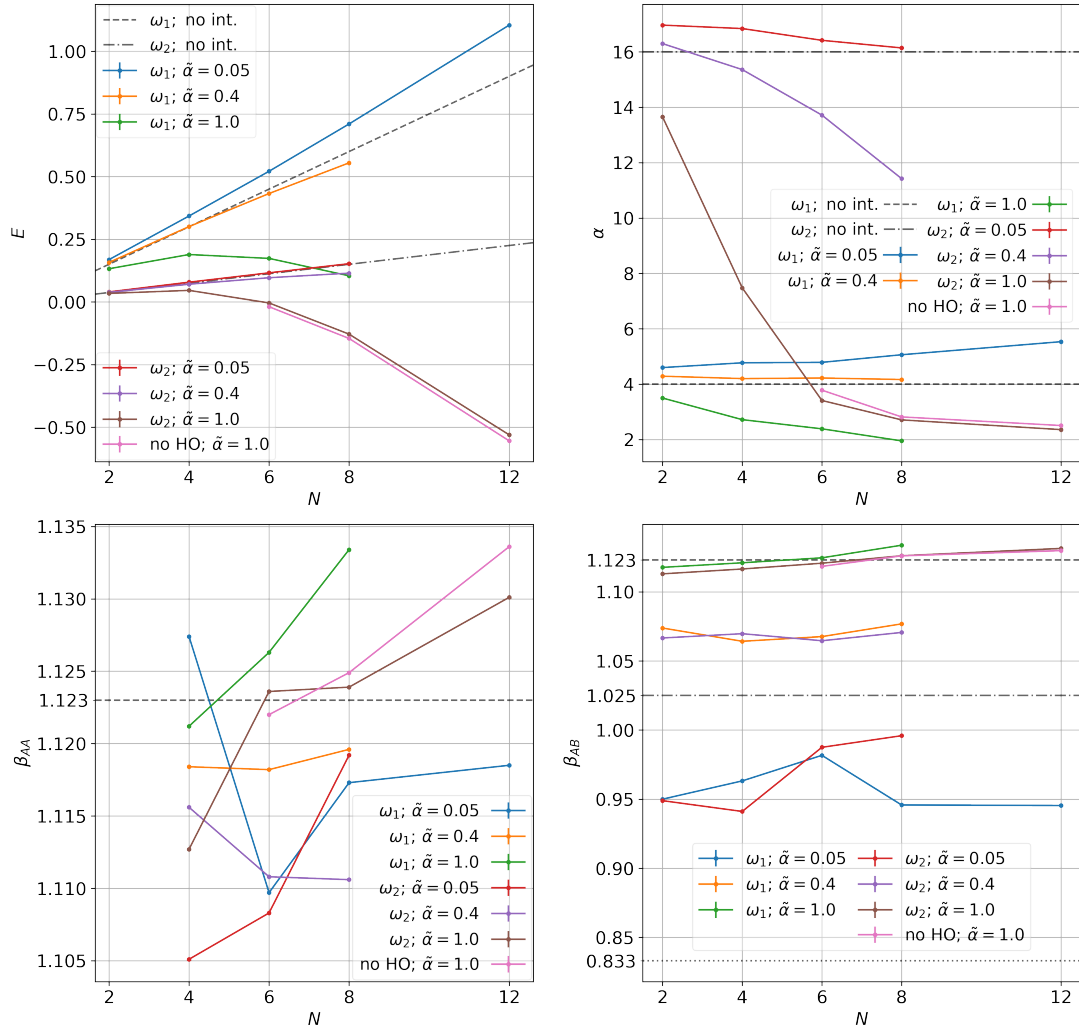


FIGURE 4.17: Summary of the results obtained for the interacting systems.

Conclusions

The Variational Monte Carlo method has proven to be a highly effective computational framework for investigating the complex many-body physics of the systems analyzed in this thesis. Through its robust implementation, we successfully unraveled the spatial structuring and mixing behaviors of a binary bosonic mixture. Most notably, we simulated the interaction-driven crossover from a confined trapped gas to a self-bound quantum liquid droplet, governed purely by the delicate thermodynamic balance of a Lennard-Jones potential.

This study complements recent theoretical investigations on quantum droplets—such as the 1D dipolar systems analyzed by De Palo et al. [3]—expanding the computational exploration to 3D Lennard-Jones mixtures. This was made possible by making extensive use of well-known computational techniques, such as Importance Sampling via Umrigar’s Drift [9], coupling them with automated routines to facilitate the parameter-tuning process.

This work also highlights several directions for future improvements. First, while the employed optimization algorithms represented a solid alternative to brute-force manual searches—speeding up the data collection process and laying a common standard for result comparison—they are far from fail-proof and proved to be disruptively affected by statistical noise. The program and the physical outcomes would greatly benefit from advanced, noise-resilient optimization algorithms (such as Stochastic Reconfiguration) specifically tailored for quantum Monte Carlo energy landscapes.

On a similar note, as discussed in Project [2], the most recent *state-of-the-art* VMC codes exploit Neural Quantum States in place of fixed Jastrow-based wavefunctions. As we demonstrated in the fully-attractive regime, a simple global Gaussian fails to naturally represent the flat-top saturation density profiles characteristic of incompressible liquids. Neural networks bypass this limitation, opening up an unprecedented level of flexibility to represent the true ground-state wavefunction and fully taking advantage of the Variational Principle. Finally, the same physical system could be investigated from entirely different perspectives. For instance, one could improve the ground-state accuracy by employing Diffusion Monte Carlo (DMC) techniques, which project out the exact ground-state employing imaginary-time evolution. Alternatively, one could simulate the real-time dynamics of the droplets following a sudden trap release to study non-equilibrium fragmentation [3], or alter the system radically by replacing bosons with fermions, thus comparing the fundamentally different behaviors dictated by the Pauli exclusion principle in totally anti-symmetric systems.

CONCLUSIONS

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